ROSE5730 Autonomous Mobile Robots

Lecture 4: Perception



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Outline

Perception

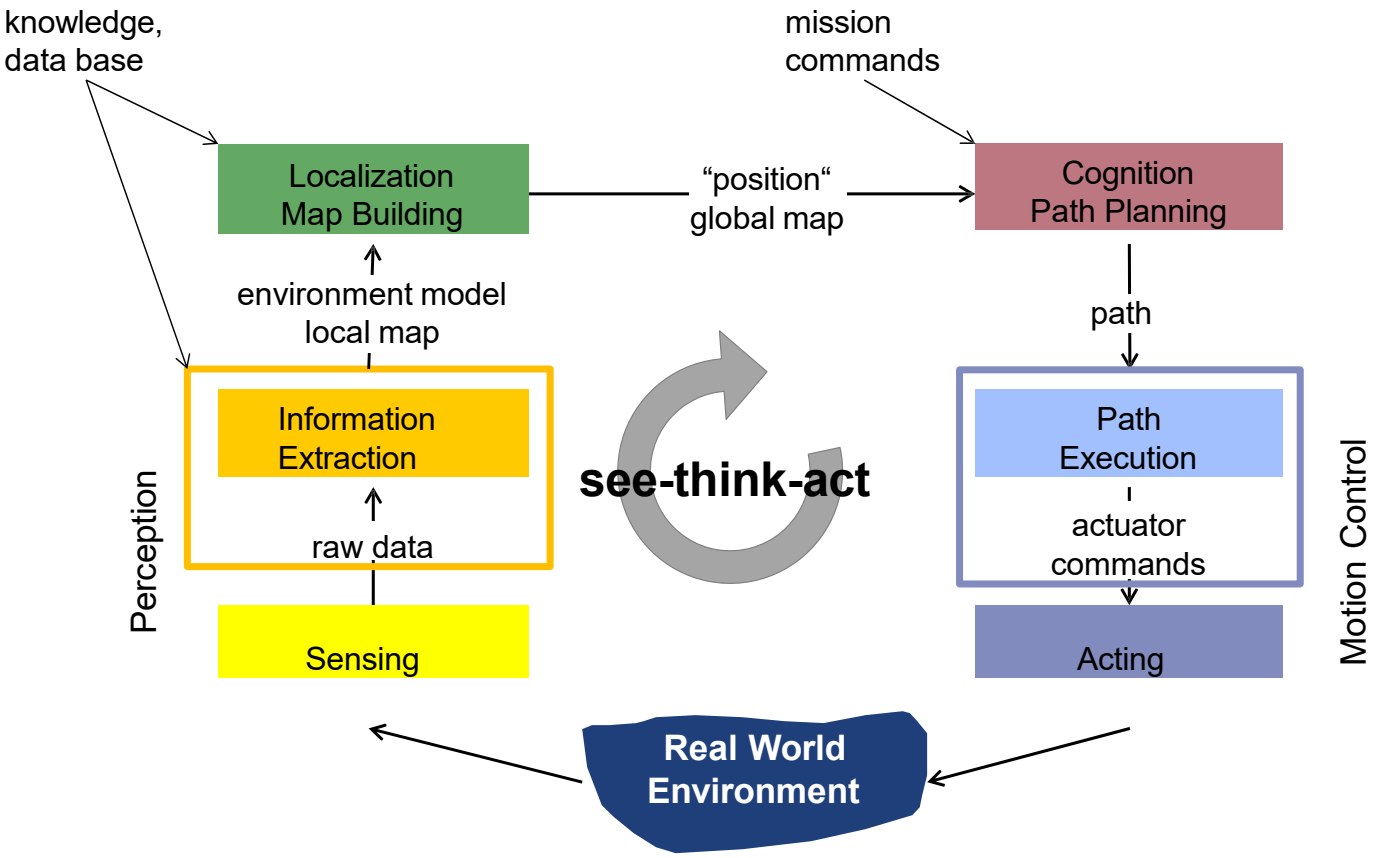
3 Stereo vision

2 Perspective projection

1 Sensors for mobile robot

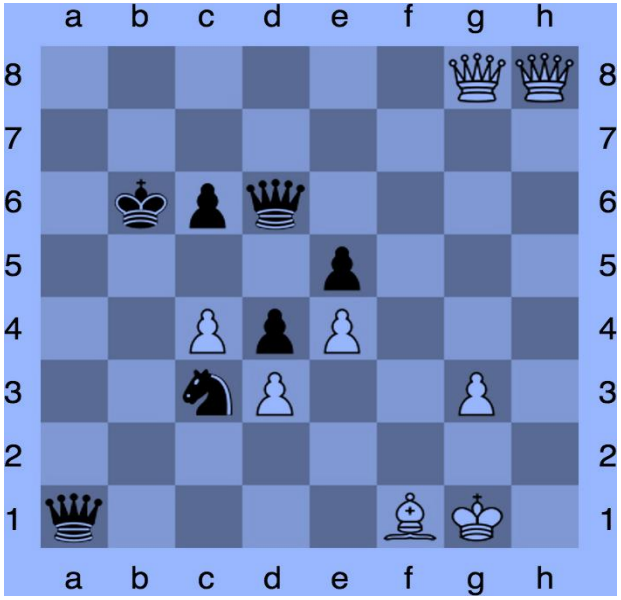


Mobile robot control scheme



Perception is hard

- "In AI, the easy problems are hard, and the hard problems are easy". S. Pinker. The Language Instinct. New York: Harper Perennial Modern Classics, 1994



Beating the world's chess master: EASY



Create a machine with some "common sense": very HARD

Sensor classification

- **Proprioceptive sensors** measure values internal to the system (robot), for example, motor speed, wheel load, robot arm joint angles, and battery voltage.
- **Exteroceptive sensors** acquire information from the robot's environment, for example distance measurements, light intensity, and sound amplitude.
- **Passive sensors** measure ambient environmental energy entering the sensor, for example, temperature probes, and microphones.
- **Active sensors** emit energy into the environment, then measure the environmental reaction.
  - They usually achieve superior performance
  - Outbound energy may affect the very characteristics that the sensor is attempting to measure. They may suffer from interference between its signal and those beyond its control.

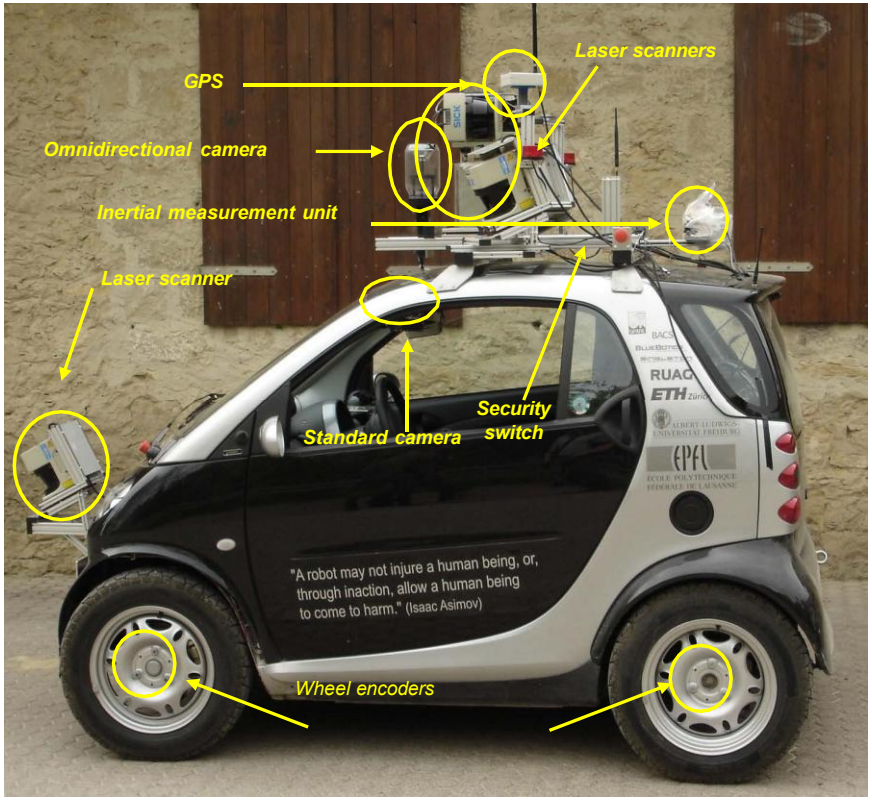
Classification

General classification (typical use)	Sensor Sensor System	PC or EC	A or P
Tactile sensors (detection of physical contact or closeness; security switches)	Contact switches, bumpers	EC	P
	Optical barriers	EC	A
	Noncontact proximity sensors	EC	A
Wheel/motor sensors (wheel/motor speed and position)	Brush encoders	PC	P
	Potentiometers	PC	P
	Synchros, resolvers	PC	A
	Optical encoders	PC	A
	Magnetic encoders	PC	A
	Inductive encoders	PC	A
	Capacitive encoders	PC	A
Heading sensors (orientation of the robot in relation to a fixed reference frame)	Compass	EC	P
	Gyroscopes	PC	P
	Inclinometers	EC	A/P
Acceleration sensor	Accelerometer	PC	P
Ground beacons (localization in a fixed reference frame)	GPS	EC	A
	Active optical or RF beacons	EC	A
	Active ultrasonic beacons	EC	A
	Reflective beacons	EC	A
Active ranging (reflectivity, time-of-flight, and geo-metric triangulation)	Reflectivity sensors	EC	A
	Ultrasonic sensor	EC	A
	Laser rangefinder	EC	A
	Optical triangulation (1D)	EC	A
	Structured light (2D)	EC	A
Motion/speed sensors (speed relative to fixed or moving objects)	Doppler radar	EC	A
	Doppler sound	EC	A
Vision sensors (visual ranging, whole-image analy-sis, segmentation, object recognition)	CCD/CMOS camera(s)	EC	P
	Visual ranging packages		
	Object tracking packages		

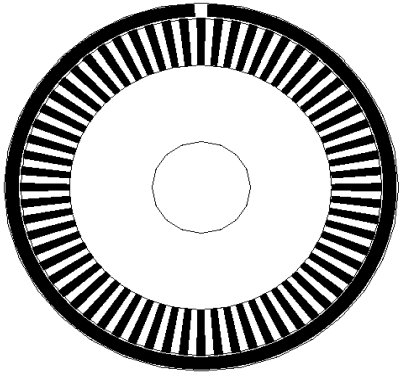
A, active; P, passive; P/A, passive/active; PC, proprioceptive; EC, exteroceptive.

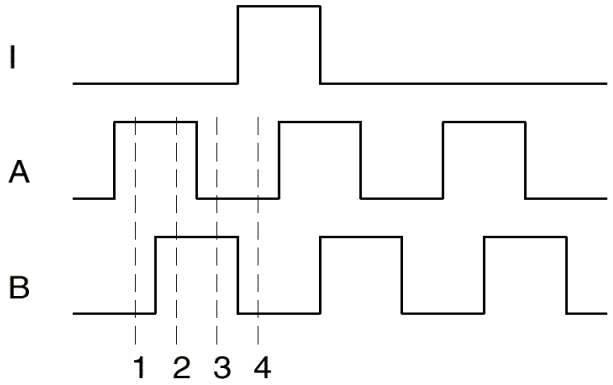
Common sensor in mobile robots

- Tactile sensors or bumpers. Detection of physical contact, security switches
- GPS: Global localization and navigation
- Inertial Measurement Unit (IMU). Orientation and acceleration of the robot
- Wheel encoders. Local motion estimation (odometry)
- Laser scanners. Obstacle avoidance, motion estimation, scene interpretation (road detection, pedestrians)
- Cameras. Texture information, motion estimation, scene interpretation



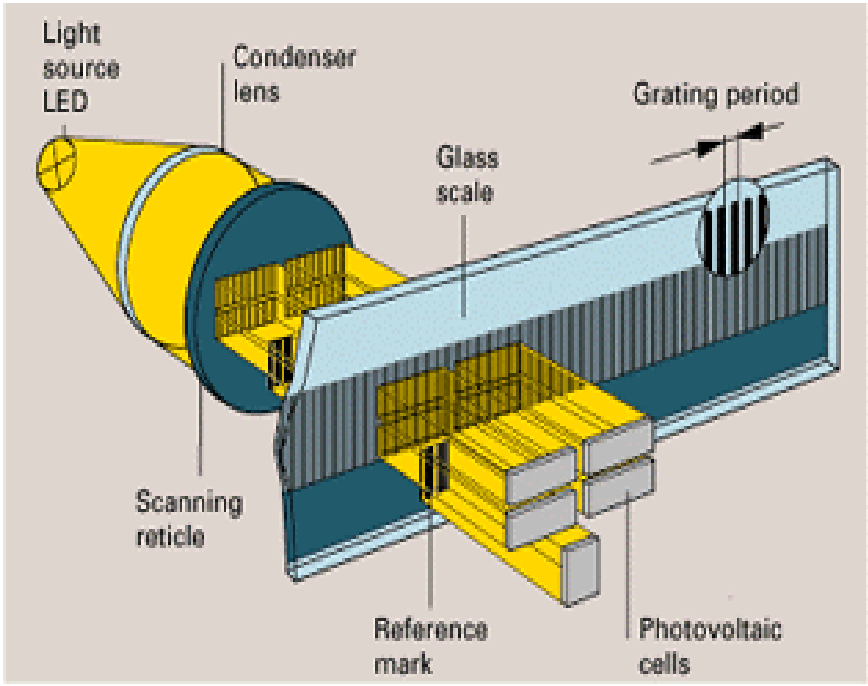
Wheeled motor encoders



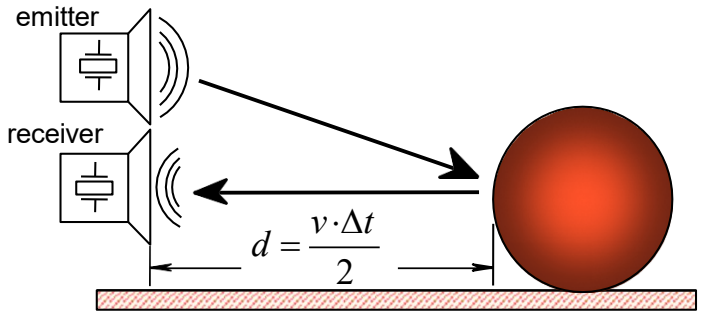


State	Ch A	Ch B
S_1	High	Low
S_2	High	High
S_3	Low	High
S_4	Low	Low

- Measure position or speed of the wheels or steering
- Integrate wheel movements to get an estimate of the position → odometry
- Optical encoders are proprioceptive sensors
- Typical resolutions: 64 - 2048 increments per revolution.
- For high resolution: interpolation



Ultrasonic Range Sensor





<http://www.robot-electronics.co.uk/>

Operational Principle

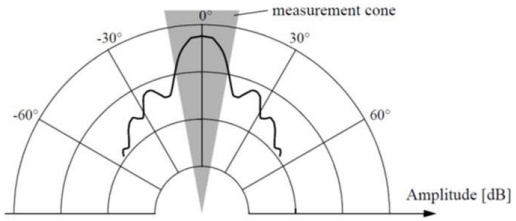
An ultrasonic pulse is generated by a piezo- electric emitter, reflected by an object in its path, and sensed by a piezo-electric receiver. Based on the speed of sound in air and the elapsed time from emission to reception, the distance between the sensor and the object is easily calculated.

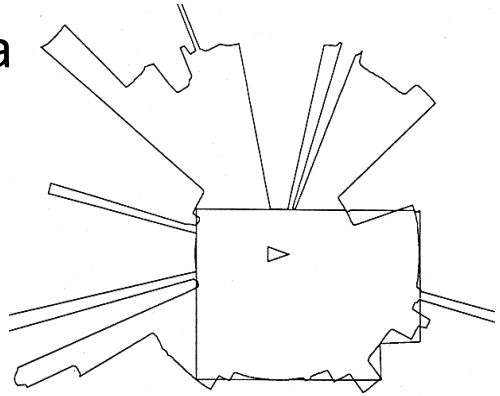
Main Characteristics

- Precision influenced by angle to object (as illustrated on the next slide)
- Useful in ranges from several cm to several meters
- Typically relatively inexpensive

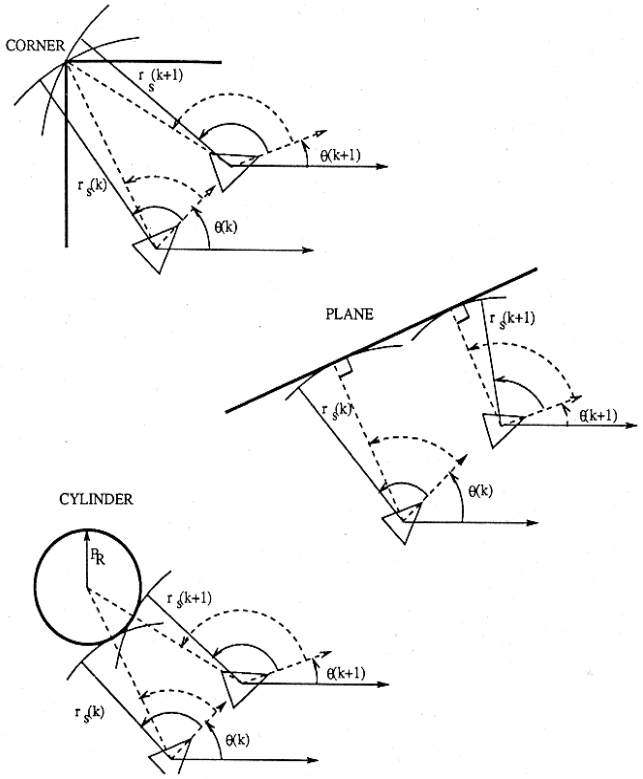
Limitations of Ultrasonic Range Sensor

- not directive – opening cone
- soft surfaces that absorb most of the sound energy
- surfaces that are rare from being perpendicula to the direction of the sound
- specular reflections
- Low update frequency (speed of sound)



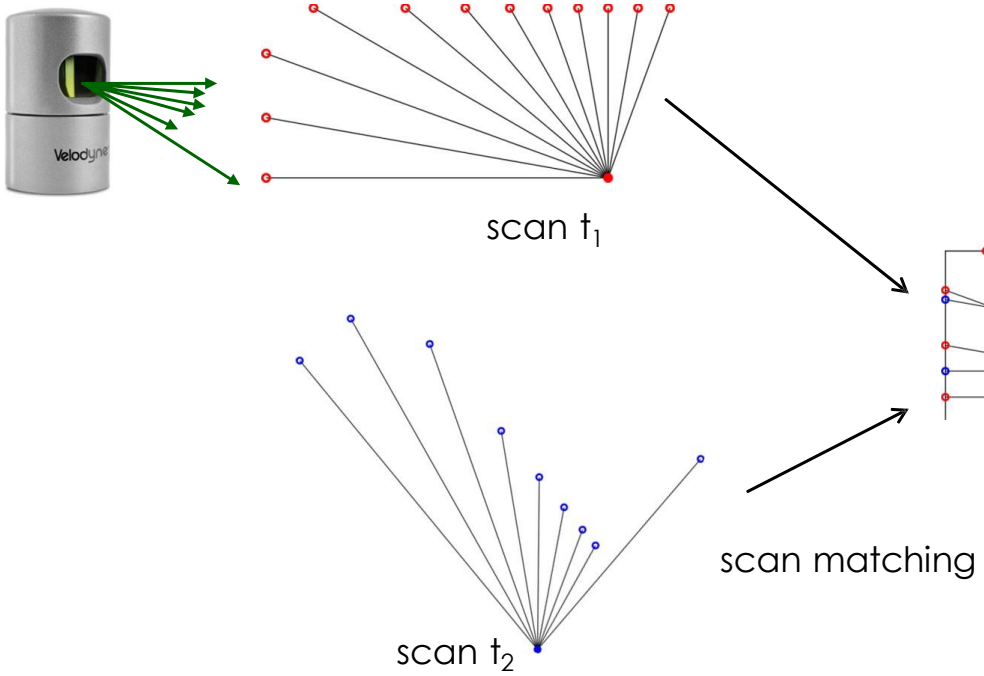


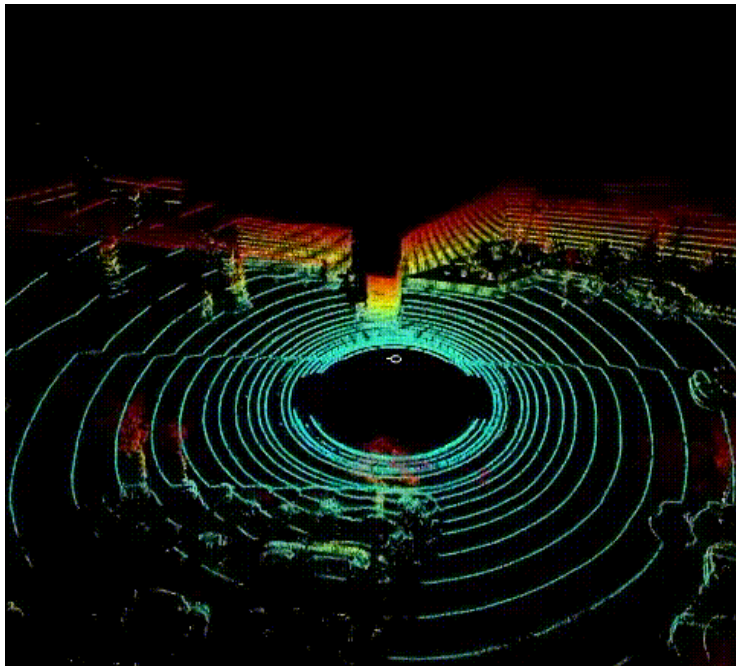
a) 360° scan



b) results from different geometric primitives

Laser-based 3D mapping

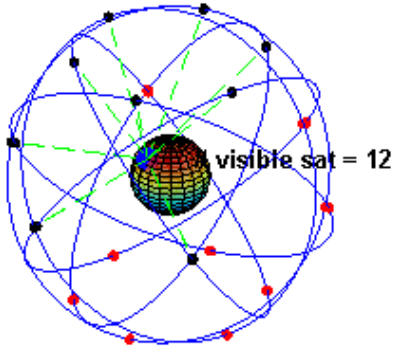




Global Positioning System (GPS)

- Working Principle
  - Location of any GPS receiver is determined through a time-of-flight measurement between the satellites and user.
  - The satellites send their orbital location (ephemeris) plus time; the receiver computes its location through trilateration and time correction.

- Technical challenges:
  - Time synchronization between the individual satellites and the GPS receiver
  - Real time update of the exact location of the satellites
  - Precise measurement of the time-of-flight
  - Interferences with other signals

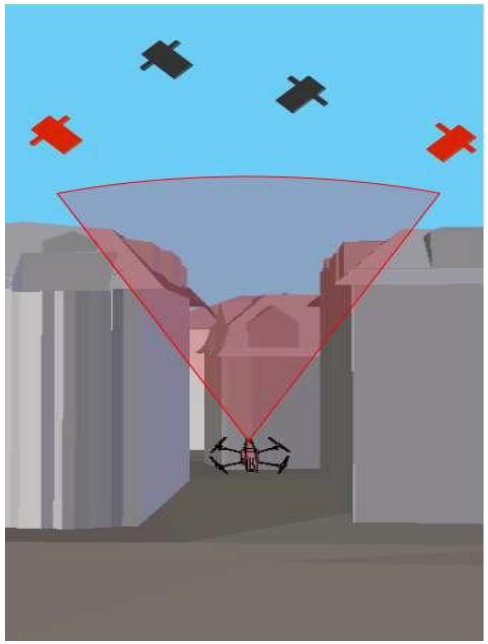


- Depending on the constellation, today an accuracy below one meter in global positioning can be reached.

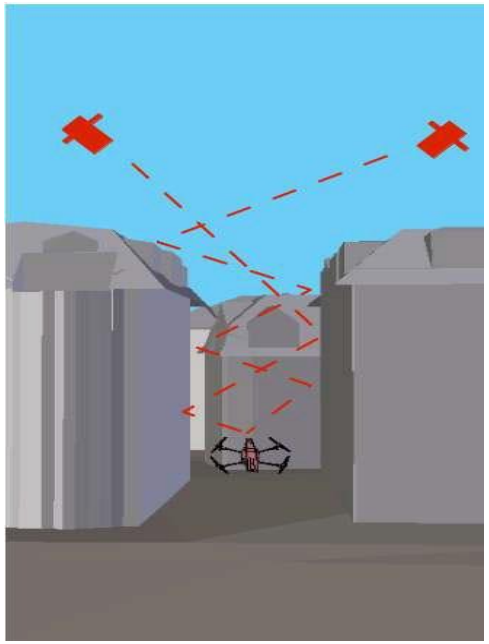
Global Positioning System (GPS)

- Interference with local environment

Satellite coverage



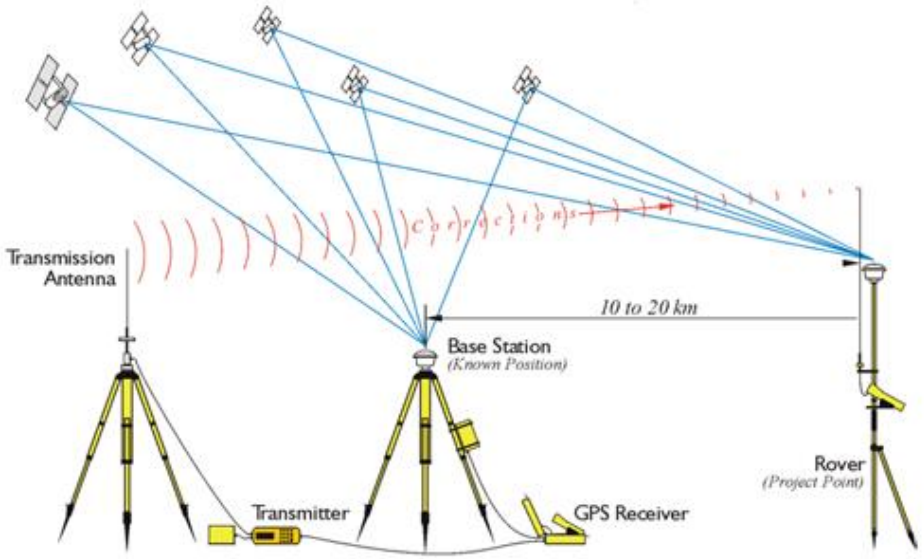
Multipath problem







Global Positioning System (GPS)

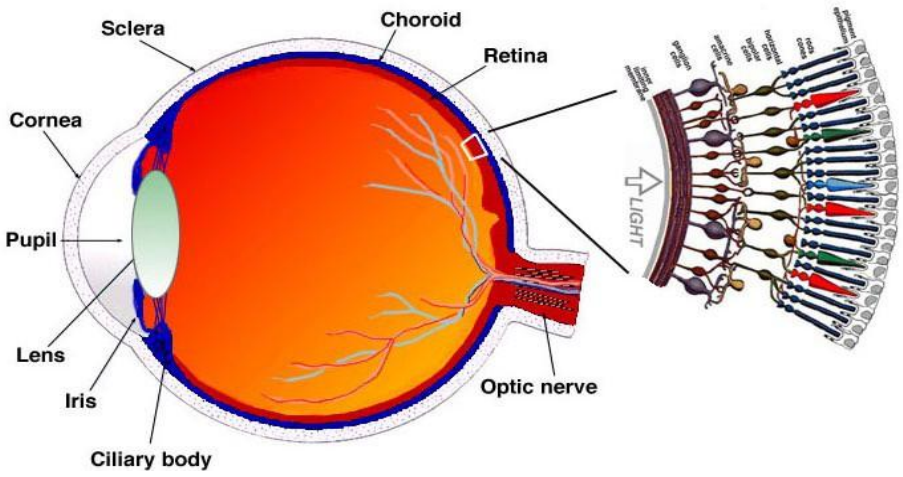


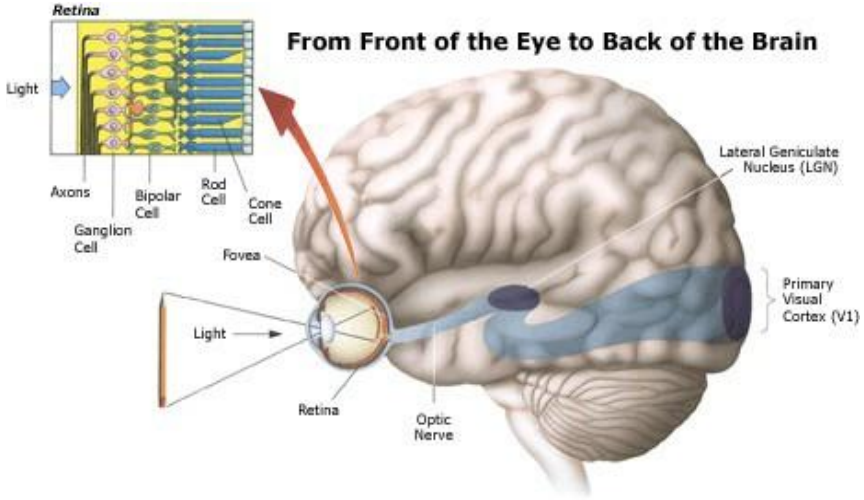
- Standard GPS
  - Ephemeris (satellite position) errors: 1 m
  - Tropospheric delays: 1 meter
  - Unmodeled ionosphere delays: 10 met
  - Multipath: 0.5 - 100 meters
  - Number of satellites under line of sight

- RTK (Real-Time Kinematic) - GPS
  - Additional measurements of the phase signal's carrier wave
  - reference station or interpolated virtual station to provide real-time corrections
  - providing up to **centimeter-level accuracy**

One picture, a thousand words

- Vision is our most powerful sense in aiding our perception of the 3D world around us
- Retina is ~10cm^2. Contains millions of photoreceptors (120 mil. rods and 7 mil. Cones for colour sampling)
- Provides enormous amount of information: data-rate of ~**3 GBytes/s**. A large proportion of our brain power is dedicated to processing the signals from our eyes





Human Visual Capabilities

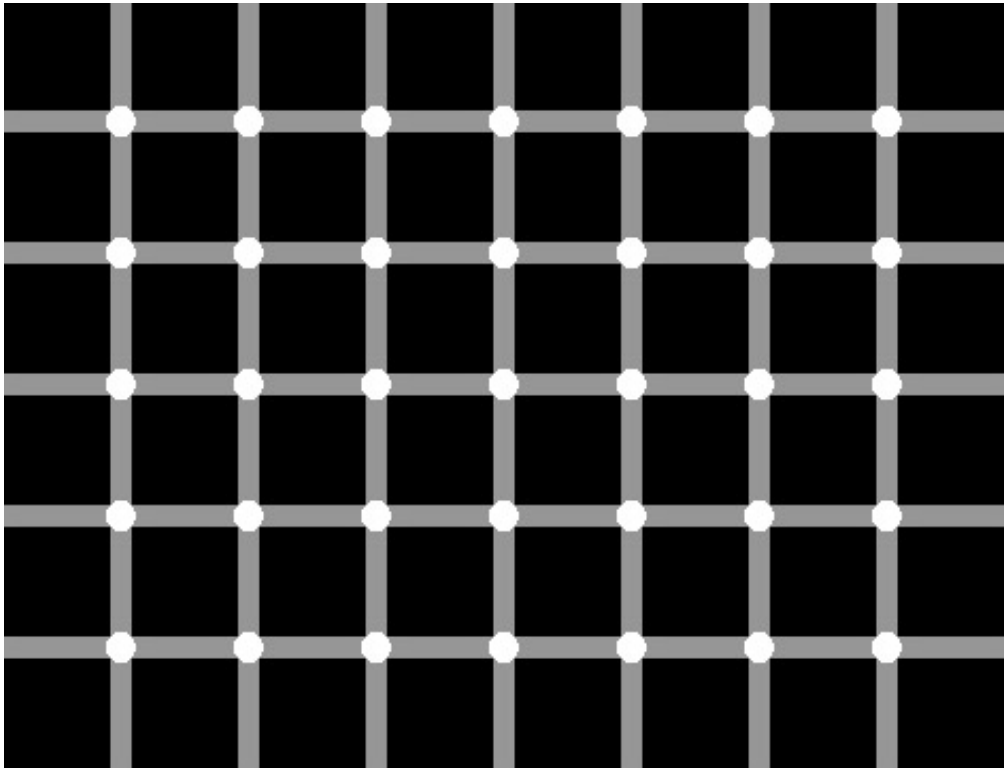
- Our visual system is very sophisticated
- Humans can interpret images successfully under a wide range of conditions – even in the presence of very limited cues





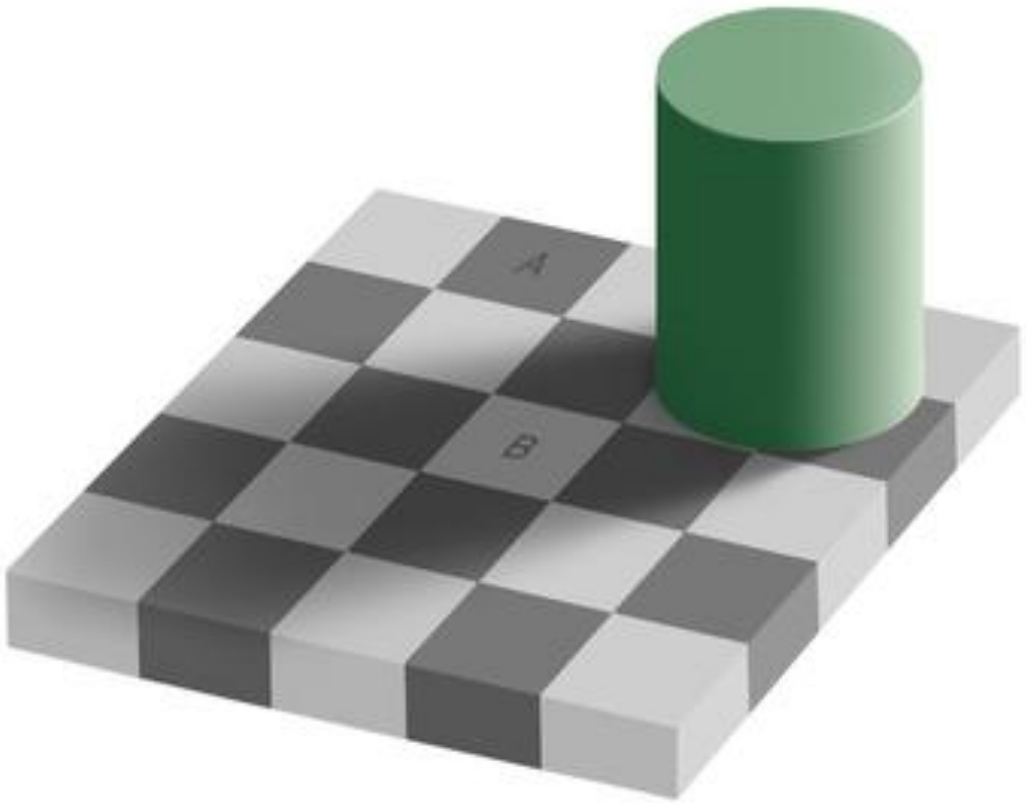
Do we get it always right?

Count the black spots



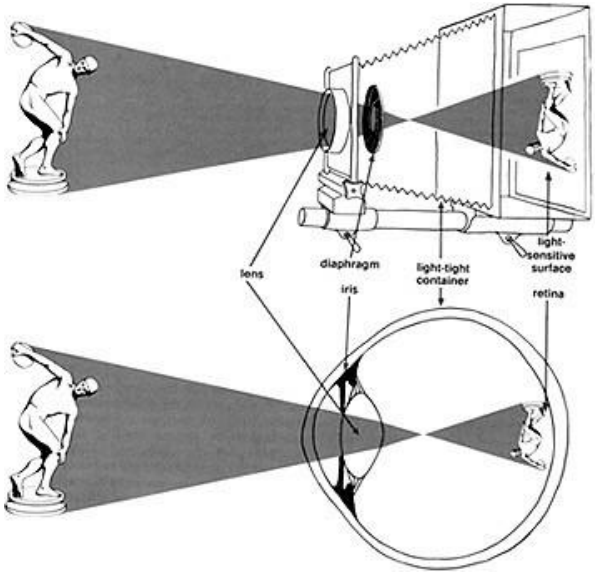
Do we get it always right?

Which square is darker, A or B?



Computer Vision for Robotics

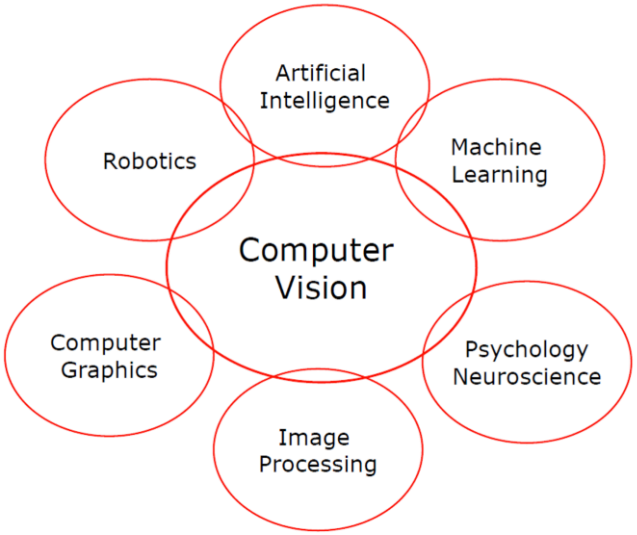
- Enormous descriptability of images → a lot of data to process (human vision involves 60 billion neurons!)
- Vision provides humans with a great deal of useful cues → explore the power of vision towards intelligent robots
- Cameras:
  - Vision is increasingly popular as a sensing modality.
  - descriptive
  - compactness, compatibility,
  - low cost
  - Hardware advances necessary to support the processing of images



Application of computer vision

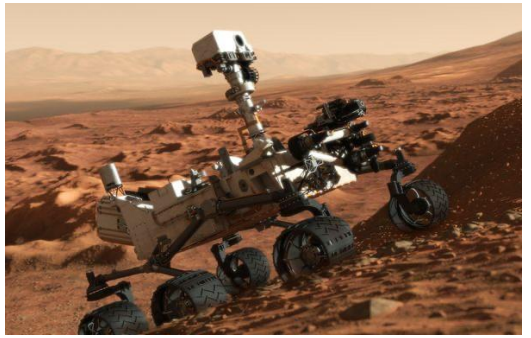






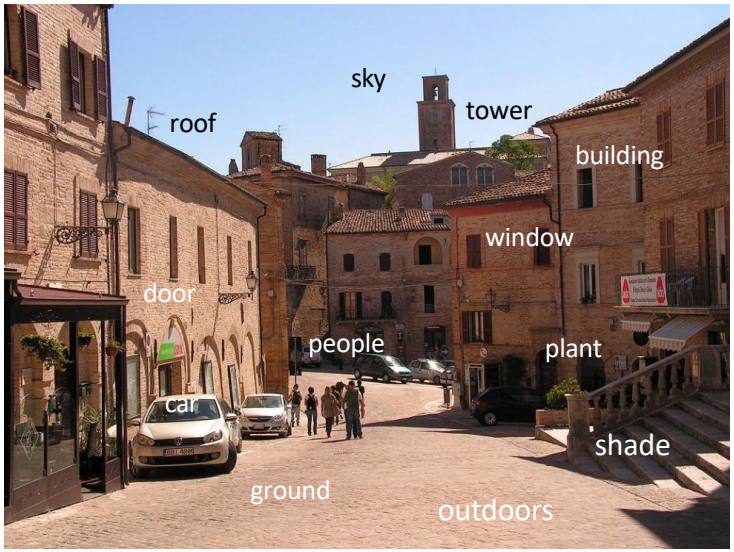




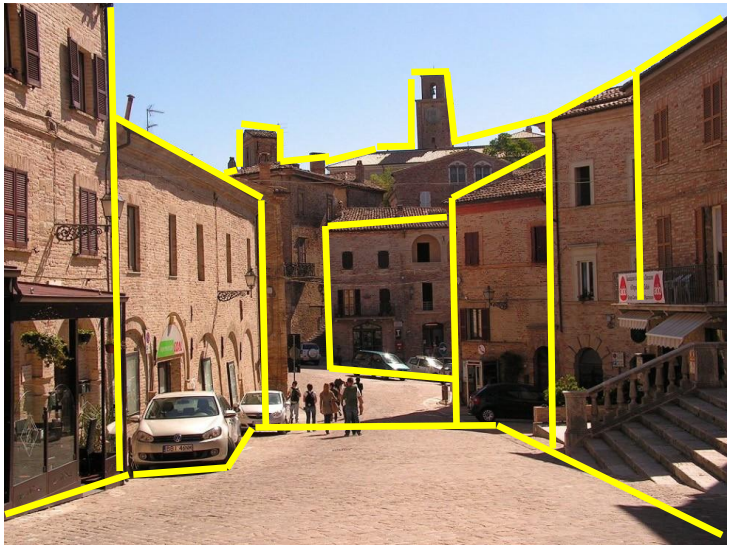


What is computer vision?

- Automatic extraction of **“meaningful” information** from images and videos – varies depending on the application



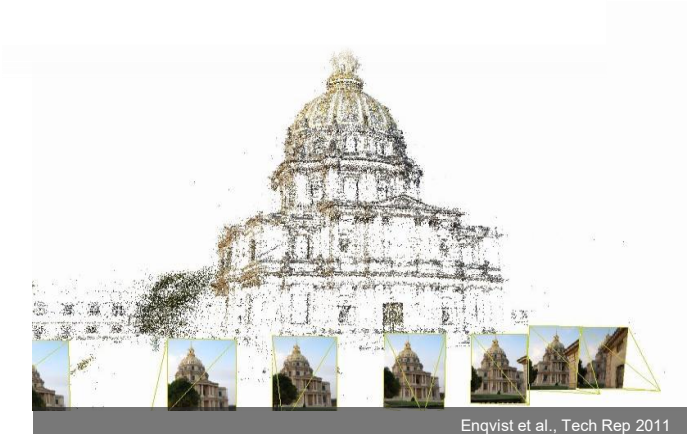
Semantic information



Geometric information

What can computer vision do?

- Partial 3D model from overlapping images



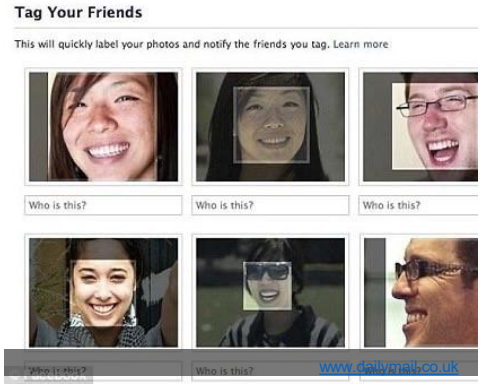
Enqvist et al., Tech Rep 2011

- Dense 3D surface model (from more overlapping images)



Goesele et al., ICCV 2007

- Face recognition



www.dailymail.co.uk



http://articles.orlandosentinel.com

- How many animals in the image?
- Resort to physics-based probabilistic models to disambiguate solutions

Image formation of a camera

If we place a piece of film in front of an object, do we get a reasonable image?

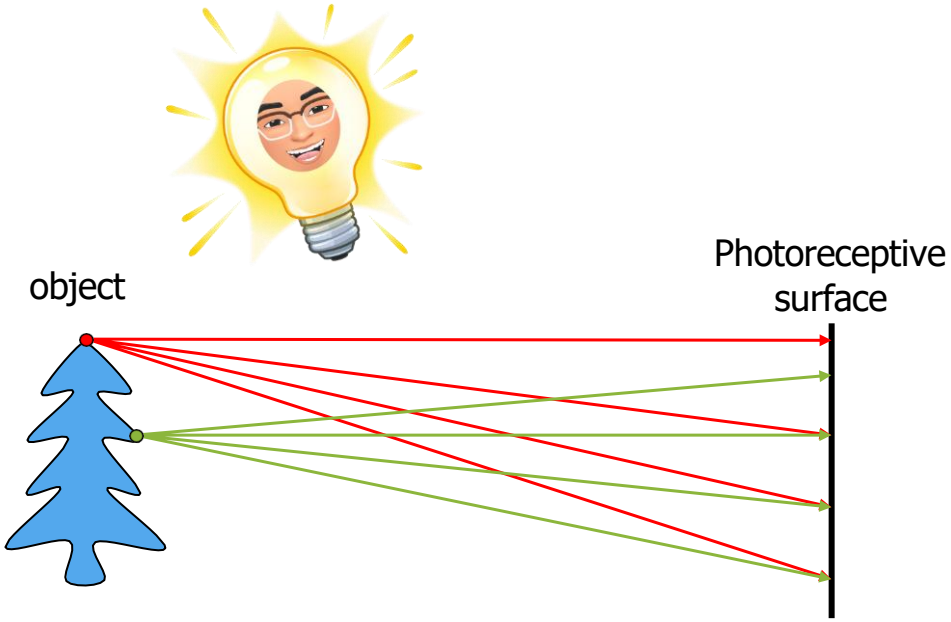
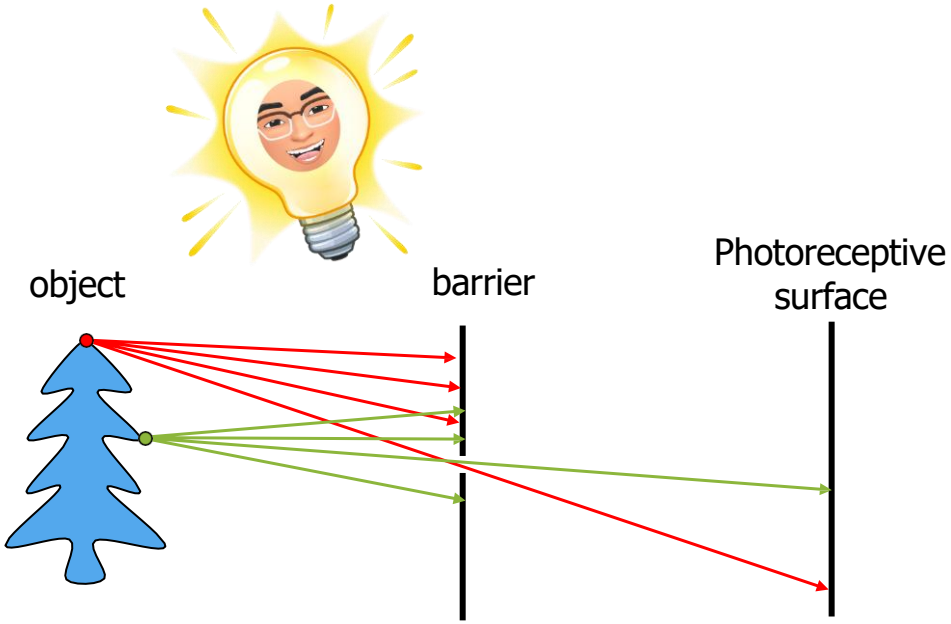
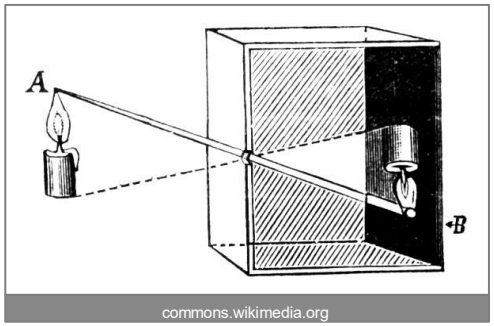


Image formation of a camera

- If we place a piece of film in front of an object, do we get a reasonable image?
- Add a barrier to block off most of the rays
- This reduces blurring
- The opening is known as the **aperture**

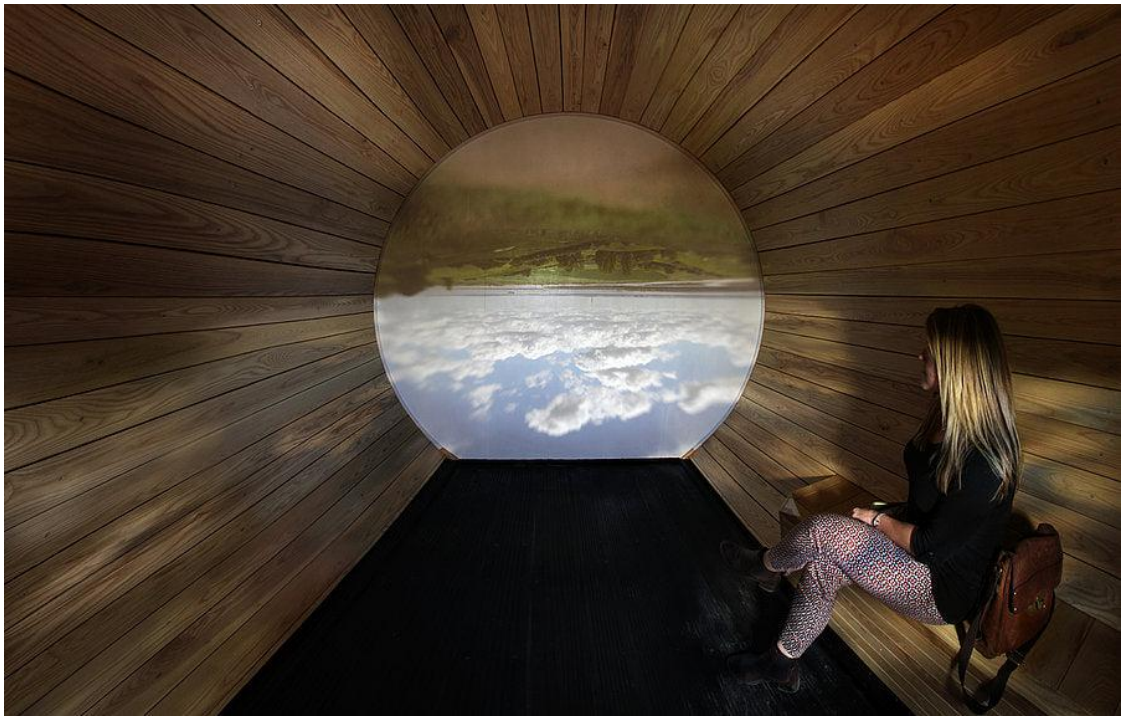


The camera obscura

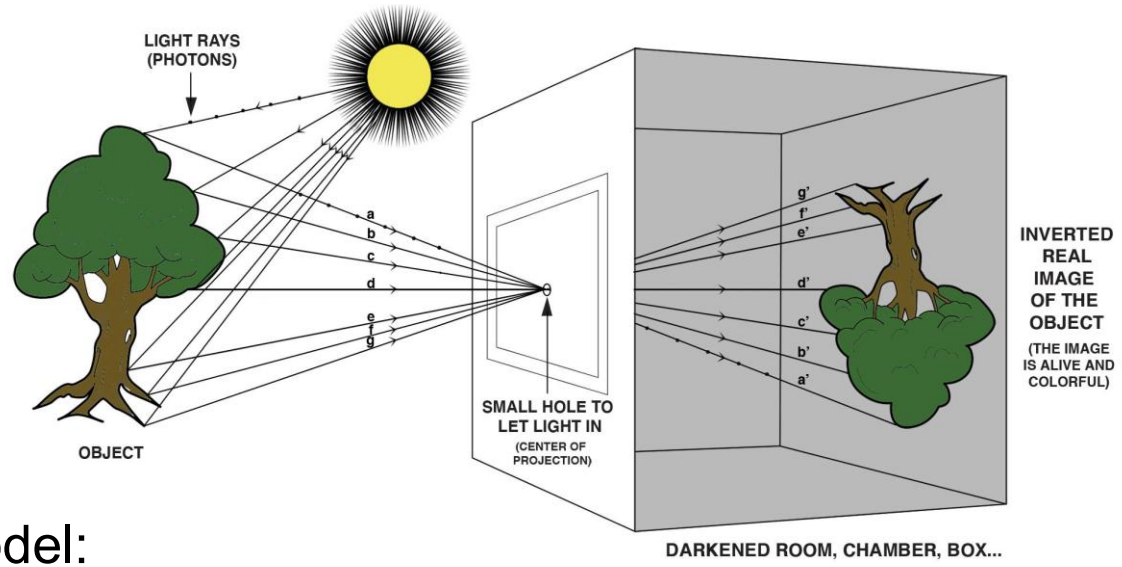


- Basic principle known to Mozi (470-390 BC), Aristotle (384-322 BC), Euclid (323-283 BC)
- Drawing aid for artists: described by Leonardo da Vinci (1452-1519)
- Image is inverted
- Depth of the room (box) is the effective focal length





The pinhole camera model



- Pinhole model:
  - Captures beam of rays – all rays through a single point (note: no lens!)
  - The point is called **Center of Projection** or **Optical Center**
  - The image is formed on the Image Plane

- We will use the pinhole camera model to describe how the image is formed

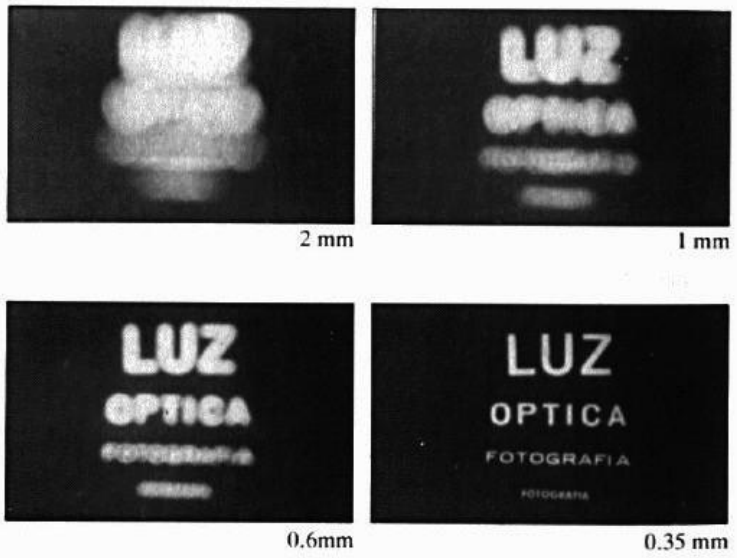
Home-made pinhole camera

Discussion: What can we do to reduce the blur?



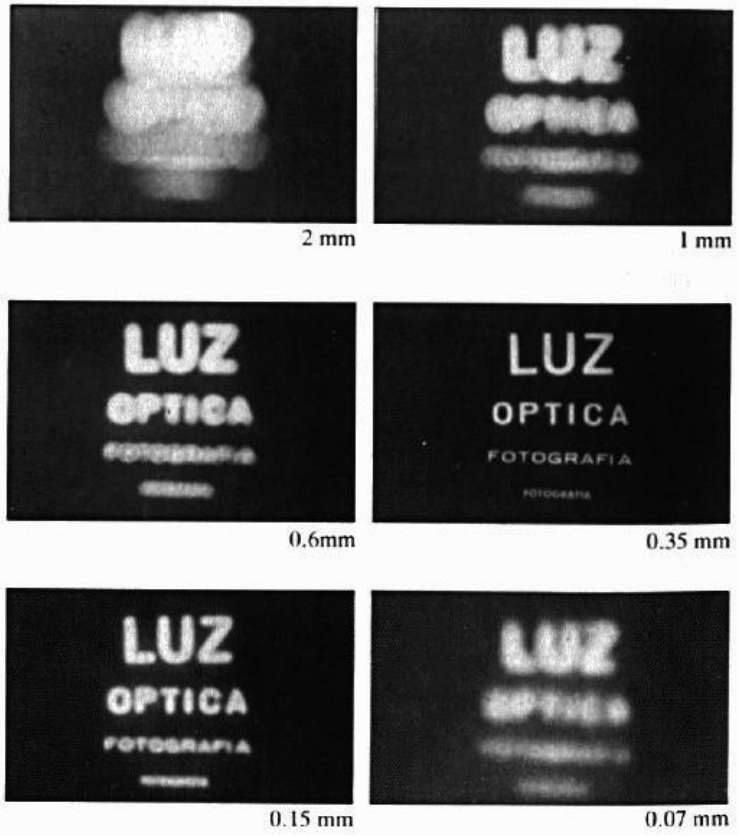


Home-made pinhole camera



Why not make the aperture as small as possible?

Shrinking the aperture

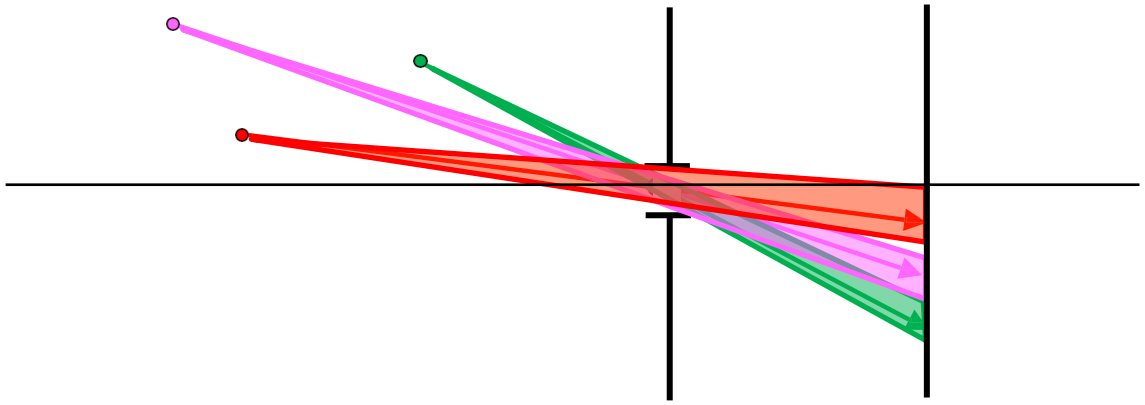


- Why not make the aperture as small as possible?
  - Less light gets through (must increase the exposure)
  - Diffraction effects…

Why use a lens?

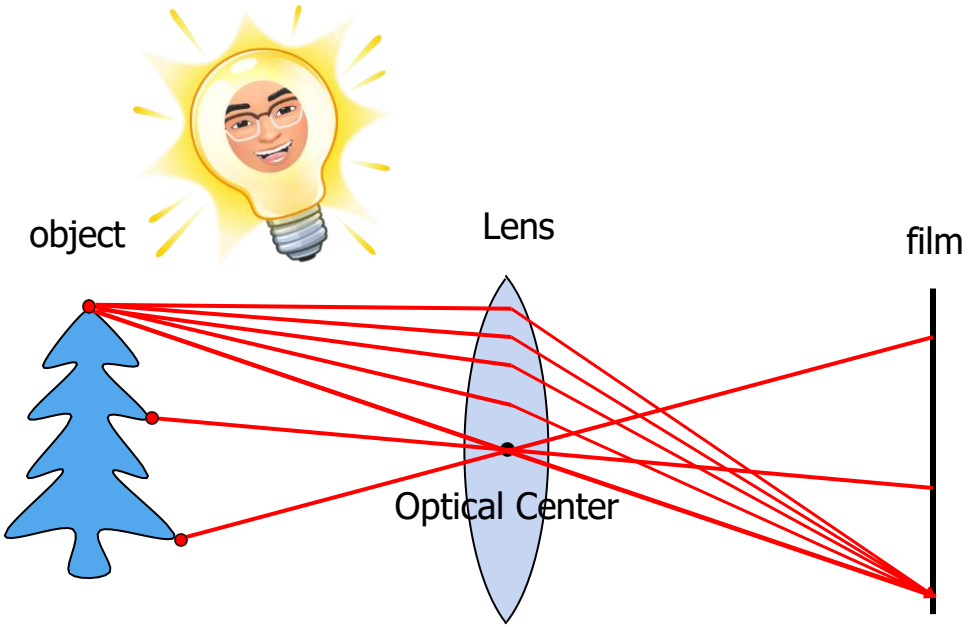
- The ideal pinhole:
  - only one ray of light reaches each point on the film
  - image can be very dim; gives rise to diffraction effects

- Making the pinhole bigger (i.e. aperture) makes the image blurry



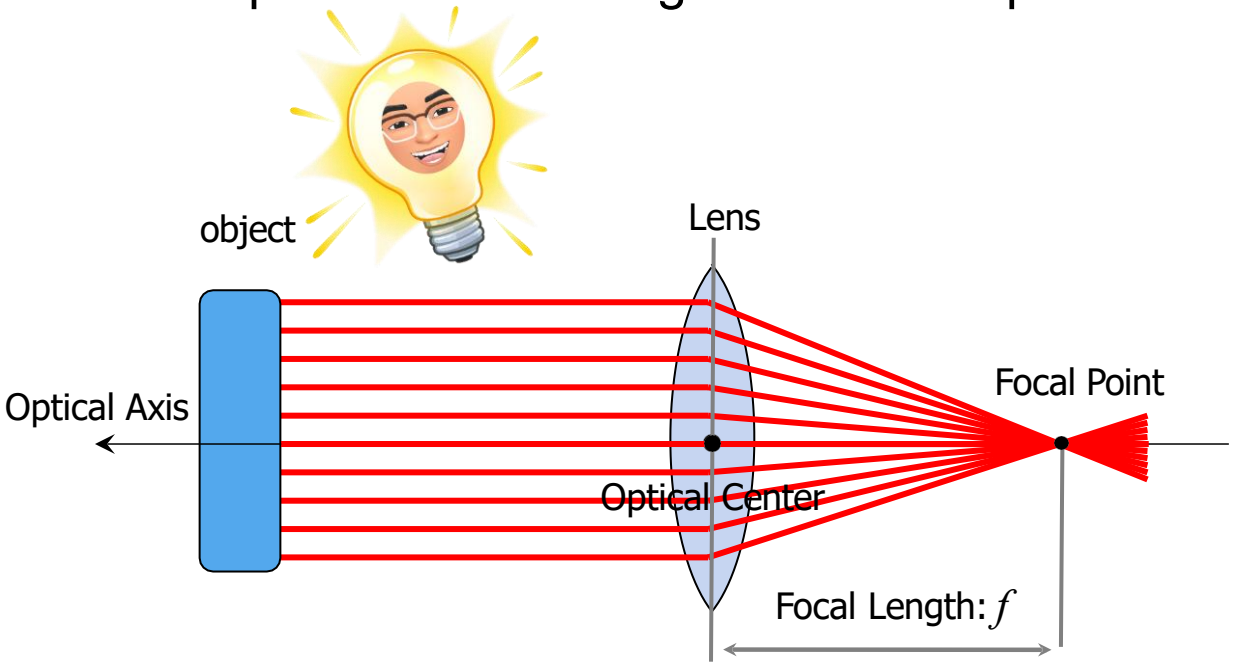
Why use a lens?

- A lens focuses light onto the film
- Rays passing through the optical center are not deviated

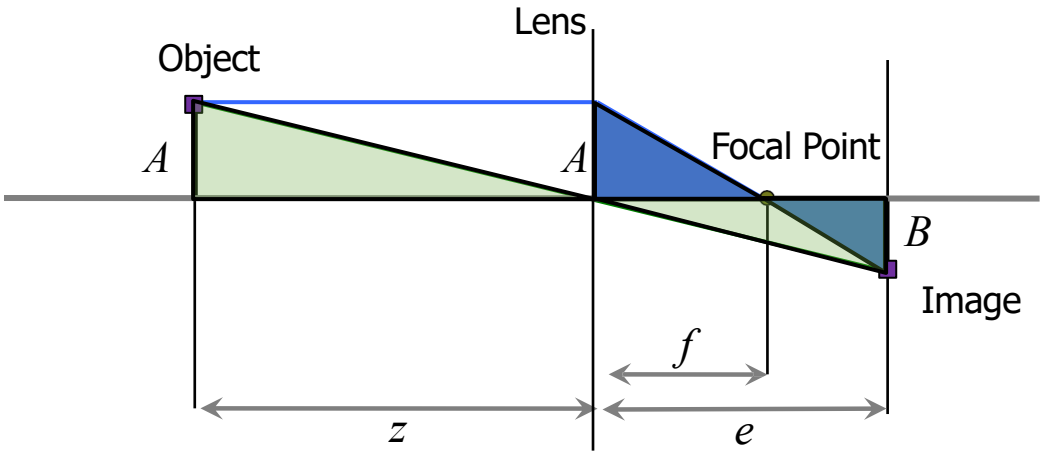


Why use a lens?

- A lens focuses light onto the film
- Rays passing through the optical center are not deviated
- All rays parallel to the optical axis converge at the focal point



The thin lens equation



- Summary of denotation:
  - z : distance to an object from lens
  - e : distance to the focal plane from lens
  - f : the focal length
  - A : Height of object
  - B : Height of image

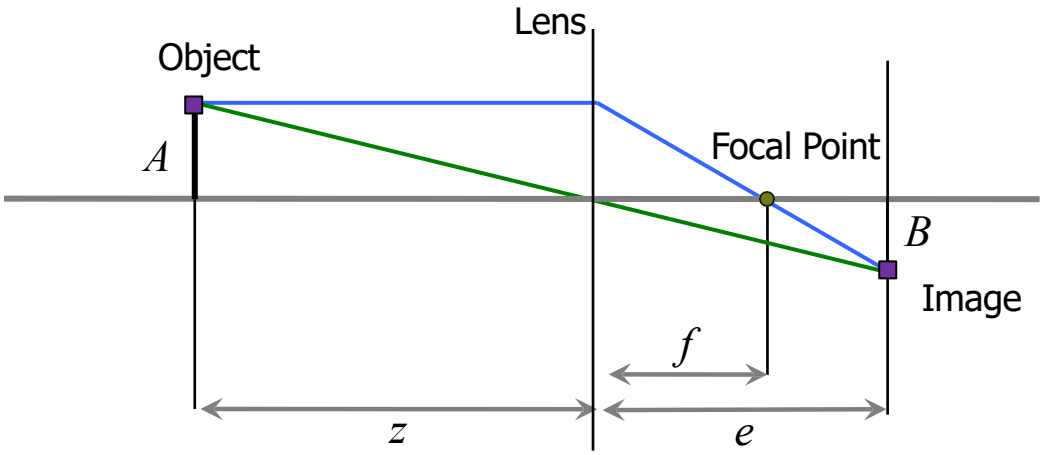
- Similar Triangles:

$$\left.\begin{aligned} \frac{B}{A} &= \frac{e}{z} \\ \frac{B}{A} &= \frac{e-f}{f} = \frac{e}{f} - 1 \end{aligned}\right\} \quad \frac{e}{f} - 1 = \frac{e}{z} \Rightarrow \boxed{\frac{1}{f} = \frac{1}{e} + \frac{1}{z}}$$

Thin lens equation



Pinhole approximation



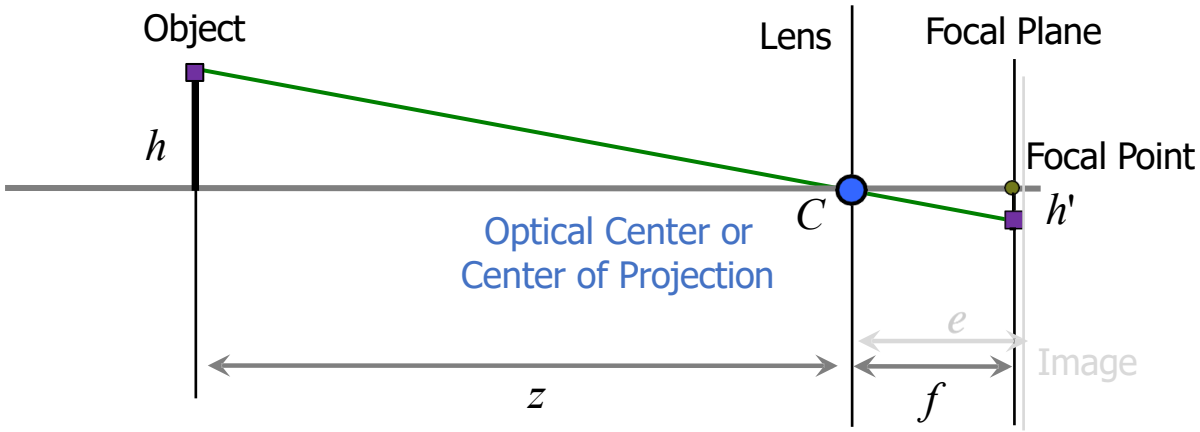
- What happens if z is very big, i.e. $z \gg f$?

$$\frac{1}{f} = \frac{1}{e} + \boxed{\frac{1}{z}}$$

$$\cong 0$$

- Summary of denotation:
  - z : distance to an object from lens
  - e : distance to the focal plane from lens
  - f : the focal length
  - A : Height of object
  - B : Height of image

Pinhole approximation



- We adjust the image plane (subject to objects at infinity are in focus)
- We can think of a lens of focal length f as being equivalent to a pinhole a distance f from the focal plane

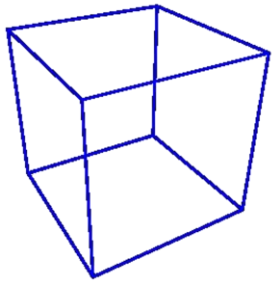
$$\frac{1}{f} \cong \frac{1}{e} \Rightarrow f \cong e$$

- The dependence of the apparent size of an object on its depth (i.e. distance from camera z) is known as ***perspective***.

$$\frac{h'}{h} = \frac{f}{z} \Rightarrow h' = \frac{f}{z} h$$

Playing with Perspective

- Perspective gives us very strong depth cues. Hence, we can perceive a 3D scene by viewing its 2D representation (i.e. image)

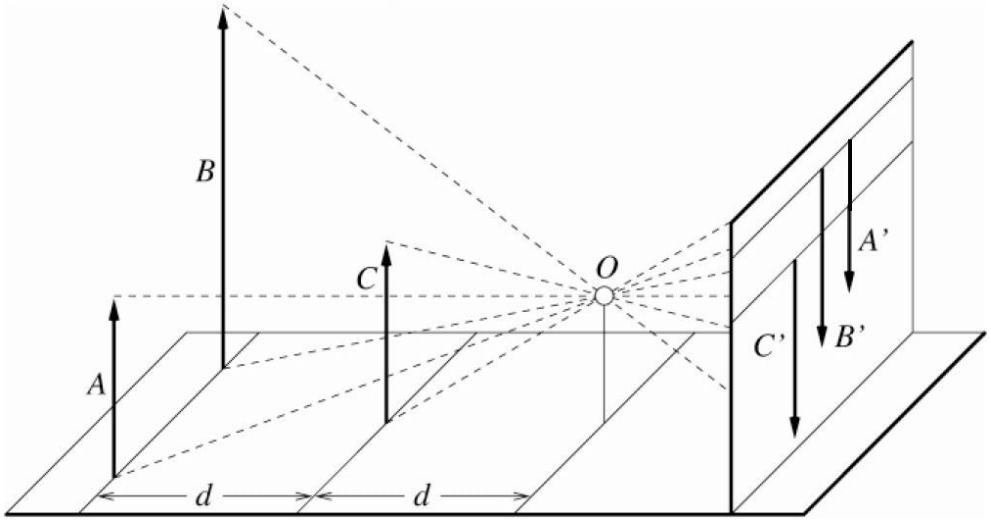




"Ames room"

Perspective effects

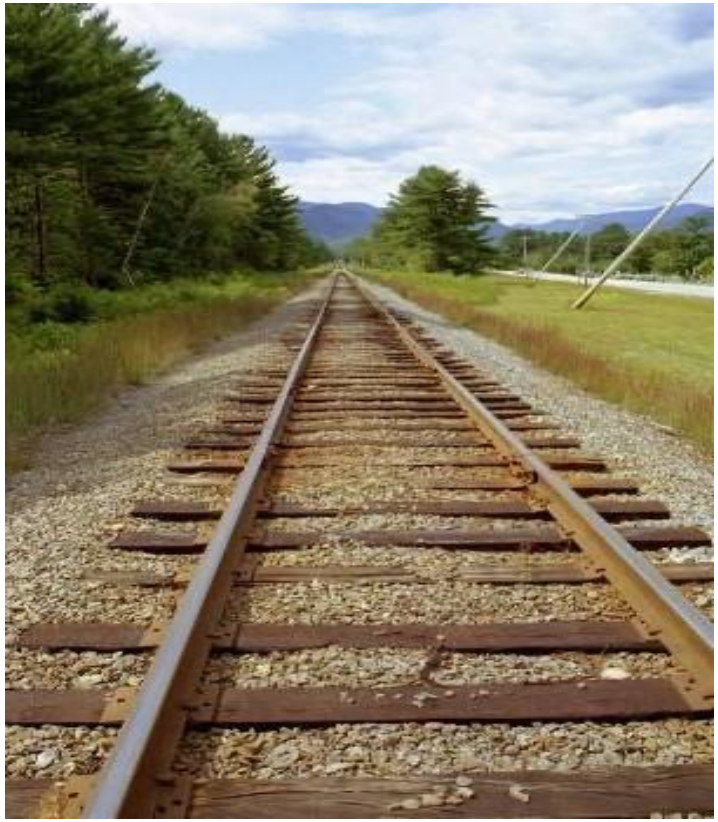
- Far away objects appear smaller



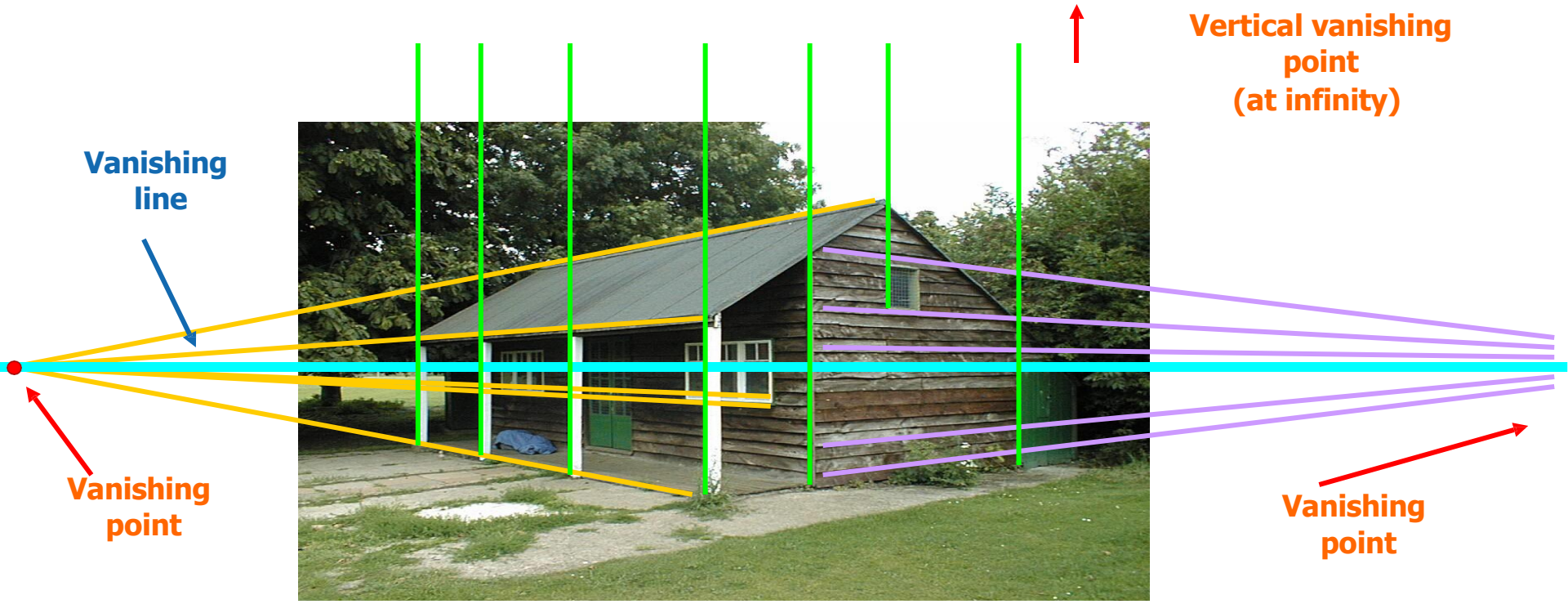


Vanishing points and lines

- Parallel lines in the world intersect in the image at a “vanishing point”



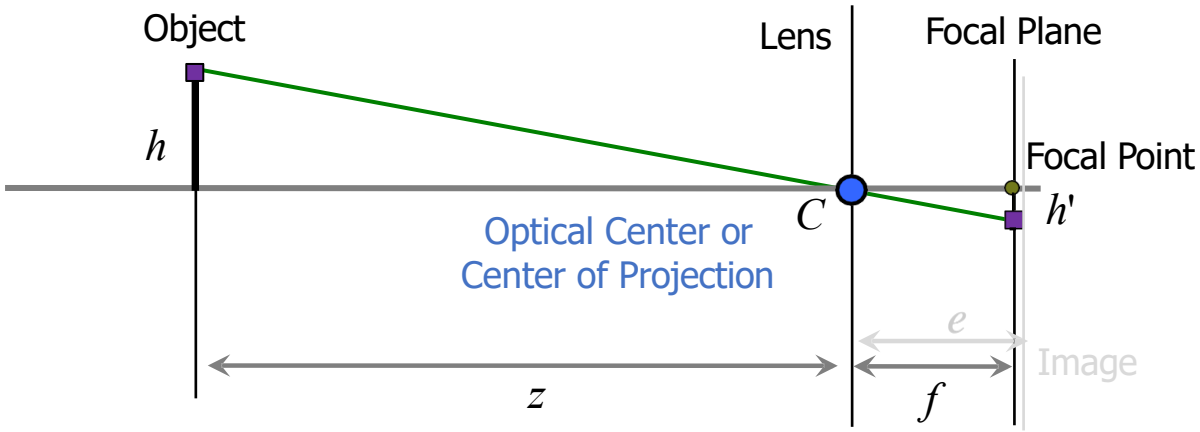
Vanishing points and lines



Vanishing points and lines



Pinhole approximation



- We adjust the image plane (subject to objects at infinity are in focus)
- We can think of a lens of focal length f as being equivalent to a pinhole a distance f from the focal plane

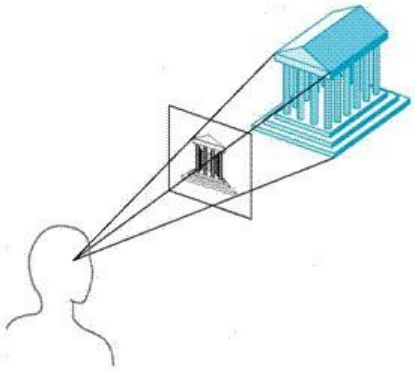
$$\frac{1}{f} \cong \frac{1}{e} \Rightarrow f \cong e$$

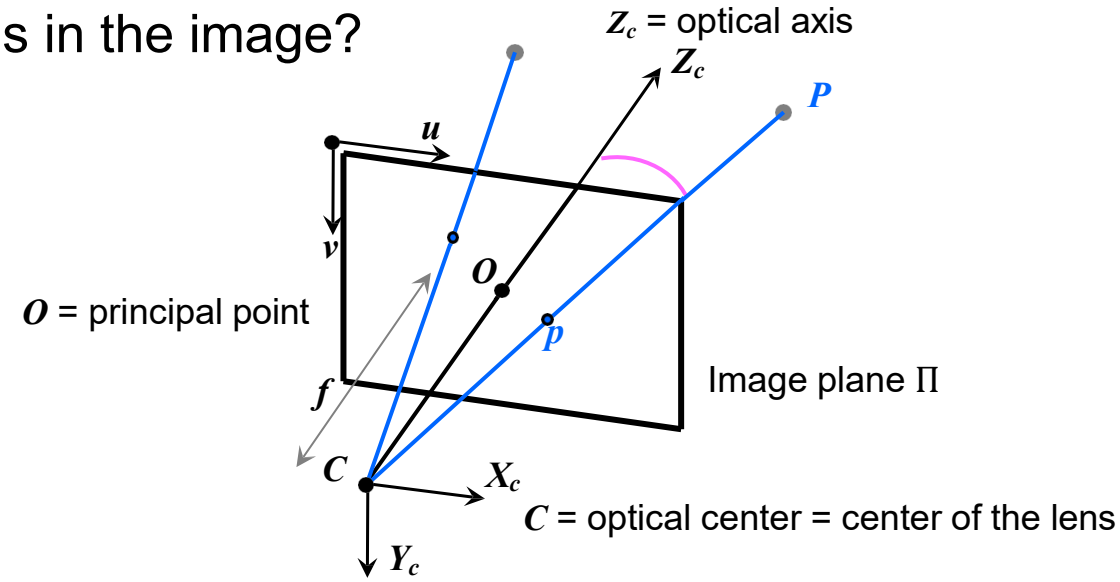
- The dependence of the apparent size of an object on its depth (i.e. distance from camera z) is known as ***perspective***.

$$\frac{h'}{h} = \frac{f}{z} \Rightarrow h' = \frac{f}{z} h$$

Perspective projection

How world points map to pixels in the image?





- Let (x, y, z) be the camera reference frame with origin in C and z-axis coincide with the optical axis Z_c.
- Introduce (u, v) a two-dimensional reference frame for the image plane with origin in O and the u and v axes aligned as x and y respectively.
- **Perspective projection**: Establish the relationship between pixel coordinates $p = (u, v)$ with 3D coordinates $P = (x, y, z)$.

From world to pixel coordinates

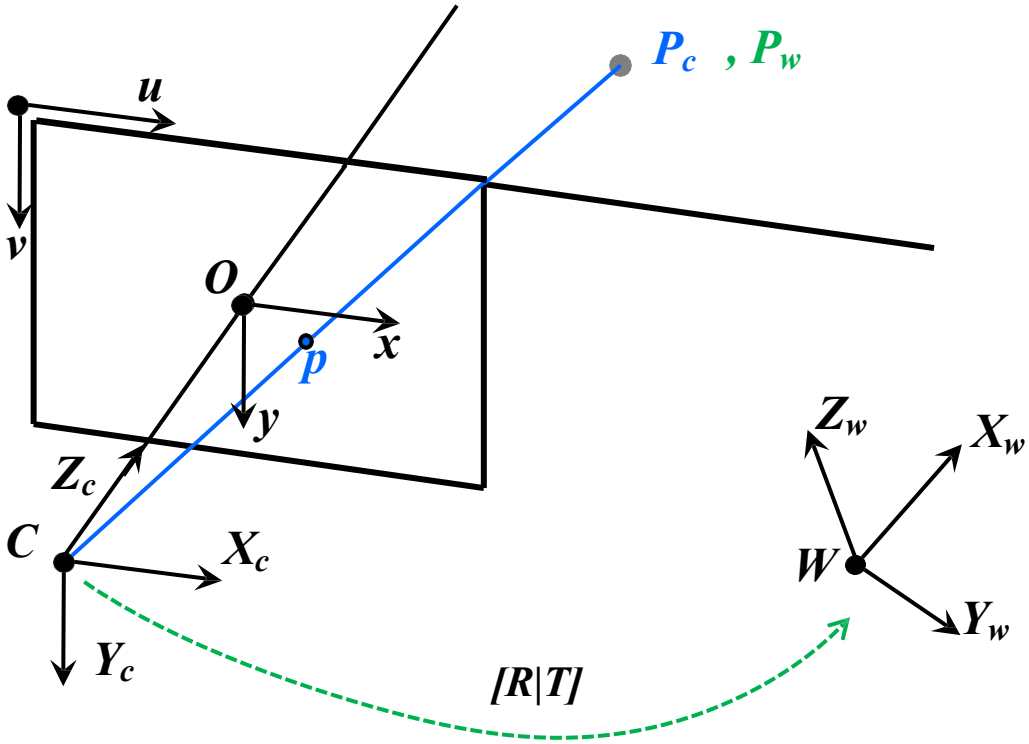
Find pixel coordinates (u, v) of point P_w in the world frame:

0. Convert world point P_w to camera point P_c

Find pixel coordinates (u, v) of point P_c in the camera frame:

1. Convert P_c to image-plane coordinates (x, y)

2. Convert P_c to (discretised) pixel coordinates (u, v)



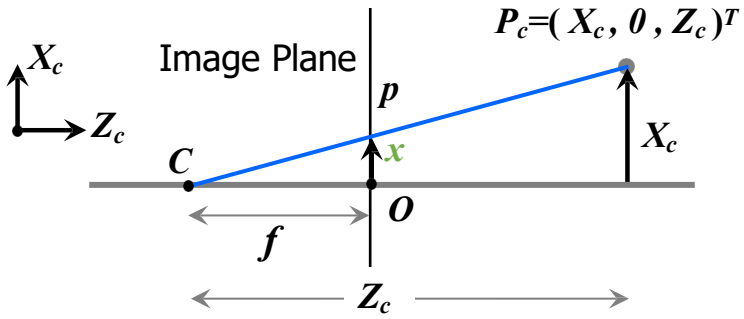
- The point P could be depicted by the camera reference coordinates $P_c = (X_c, Y_c, Z_c)$ and by the world reference coordinates $P_w = (X_w, Y_w, Z_w)$.

From camera frame to image plane

- The Camera point $P_c = (X_c, 0, Z_c)^T$ projects to $p = (x, y)$ onto the image plane

- From similar triangles:

$$\frac{x}{f} = \frac{X_c}{Z_c} \Rightarrow x = \frac{fX_c}{Z_c}$$



- Similarly, in the general case:

$$\frac{y}{f} = \frac{Y_c}{Z_c} \Rightarrow y = \frac{fY_c}{Z_c}$$

1. Convert P_c to image-plane coordinates (x, y)

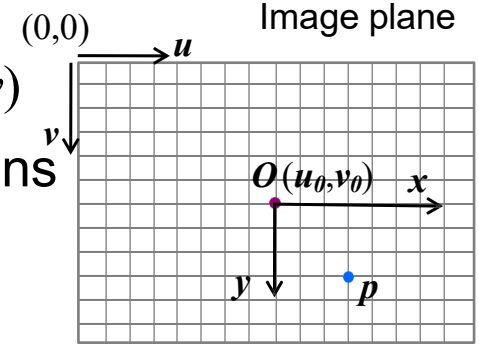
2. Convert P_c to (discretised) pixel coordinates (u, v)

from camera to pixel coordinates

- Convert p from the local image plane coords (x, y) to the pixel coords (u, v), we need to account for:

  The pixel coords of the camera optical center $O=(u, v)$

  scale factors k_u, k_v for the pixel-size in both dimensions



$$u = u_0 + k_u x \Rightarrow u = u_0 + k_u \frac{fX_c}{Z_c}$$

$$v = v_0 + k_v y \Rightarrow v = v_0 + k_v \frac{fY_c}{Z_c}$$

- Use **Homogeneous Coordinates** for linear mapping from 3D to 2D, by introducing an extra element (scale):

$$p = \begin{bmatrix} u \\ v \end{bmatrix} \Rightarrow \tilde{p} = \begin{bmatrix} \lambda u \\ \lambda v \\ \lambda \end{bmatrix} = \lambda \begin{bmatrix} u \\ v \\ 1 \end{bmatrix}$$

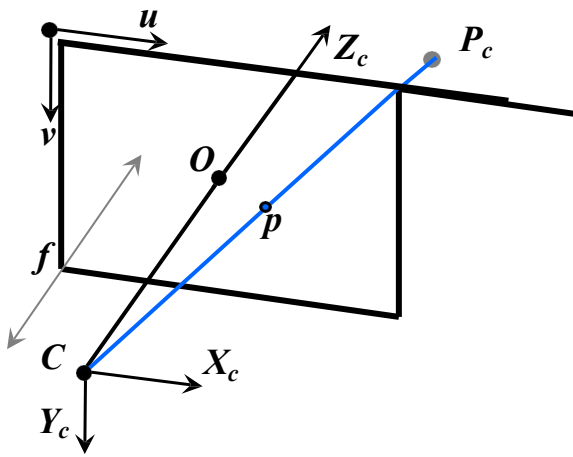
and similarly for the world coordinates.
Note, usually $\lambda = 1$

from camera to pixel coordinates

$$u = u_0 + k_u \frac{fX_c}{Z_c} \qquad v = v_0 + k_v \frac{fY_c}{Z_c}$$

- Expressed in matrix form and homogeneous coordinates:

$$\begin{bmatrix} \lambda u \\ \lambda v \\ \lambda \end{bmatrix} = \begin{bmatrix} k_u f & 0 & u_0 \\ 0 & k_v f & v_0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} X_c \\ Y_c \\ Z_c \end{bmatrix}$$



- Or alternatively

$$\begin{bmatrix} \lambda u \\ \lambda v \\ \lambda \end{bmatrix} = \begin{bmatrix} \alpha_u & 0 & u_0 \\ 0 & \alpha_v & v_0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} X_c \\ Y_c \\ Z_c \end{bmatrix} = K \begin{bmatrix} X_c \\ Y_c \\ Z_c \end{bmatrix}$$

Intrinsic parameter matrix

Focal length in v -direction

Focal length in u -direction

from world to pixel coordinates

- Let us connect the world reference coordinates with pixel coordinates

$$\begin{bmatrix} X_c \\ Y_c \\ Z_c \end{bmatrix} = \begin{bmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{bmatrix} \begin{bmatrix} X_W \\ Y_W \\ Z_W \end{bmatrix} + \begin{bmatrix} t_1 \\ t_2 \\ t_3 \end{bmatrix}$$

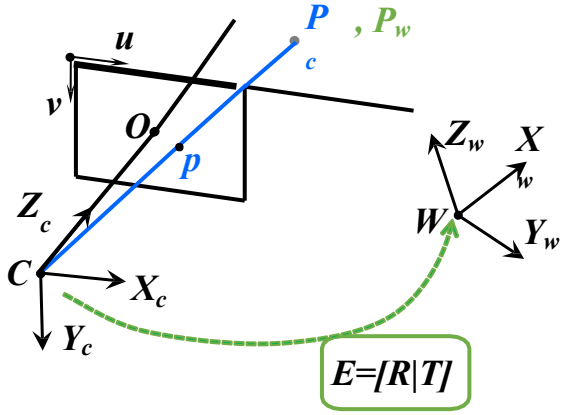
In homogenous coordinates

$$\begin{bmatrix} X_c \\ Y_c \\ Z_c \\ 1 \end{bmatrix} = \left[\begin{array}{ccc|c} r_{11} & r_{12} & r_{13} & t_1 \\ r_{21} & r_{22} & r_{23} & t_2 \\ r_{31} & r_{32} & r_{33} & t_3 \\ \hline 0 & 0 & 0 & 1 \end{array}\right] \begin{bmatrix} X_W \\ Y_W \\ Z_W \\ 1 \end{bmatrix} = \left[\begin{array}{ccc|c} & R & & \boldsymbol{t} \\ \hline 0 & 0 & 0 & 1 \end{array}\right] \begin{bmatrix} X_W \\ Y_W \\ Z_W \\ 1 \end{bmatrix}$$

- From the previous slide, one has

$$\begin{bmatrix} \lambda u \\ \lambda v \\ \lambda \end{bmatrix} = K \begin{bmatrix} X_c \\ Y_c \\ Z_c \end{bmatrix} \qquad \lambda \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = K[R \mid \boldsymbol{t}] \begin{bmatrix} X_W \\ Y_W \\ Z_W \\ 1 \end{bmatrix}$$

Projection matrix



Extrinsic Parameter matrix

Example

Given the intrinsic parameter matrix and extrinsic parameter matrix below,

$$K = \begin{bmatrix} 900 & 0 & 640 \\ 0 & 880 & 360 \\ 0 & 0 & 1 \end{bmatrix}, E = \begin{bmatrix} 0 & -1 & 0 & 0.1 \\ 1 & 0 & 0 & -0.05 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

for a world point $P_W = \begin{bmatrix} X_W \\ Y_W \\ Z_W \end{bmatrix} = \begin{bmatrix} 0.5 \\ 0.2 \\ 0 \end{bmatrix}$, please find its pixel coordinates.

From the given matrix, we can observe

$$\alpha_u = 900, \alpha_v = 880, u_0 = 640, v_0 = 360, R = \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}, t = \begin{bmatrix} 0.1 \\ -0.05 \\ 2 \end{bmatrix}$$

Example

- Solution: Step 1. World coordinates ->Camera coordinates

$$P_C = RP_W + t$$

which is $P_C = RP_W + t = \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0.5 \\ 0.2 \\ 0 \end{bmatrix} + \begin{bmatrix} 0.1 \\ -0.05 \\ 2 \end{bmatrix} = \begin{bmatrix} -0.1 \\ 0.45 \\ 2 \end{bmatrix}$

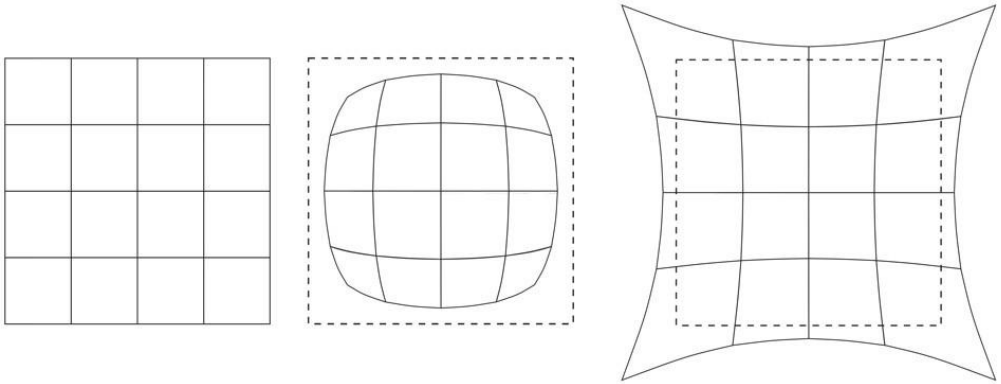
Step 2. Camera coordinates -> Pixel coordinates

$$u = u_0 + \alpha_u \frac{X_c}{Z_c} = 640 + 900 \cdot \frac{-0.1}{2} = 640 - 45 = 595 \text{ px}$$

.

$$v = v_0 + \alpha_v \frac{Y_c}{Z_c} = 360 + 880 \cdot \frac{0.45}{2} = 360 + 198 = 558 \text{ px}$$

Radial distortion





No distortion Barrel distortion Pincushion distortion

- Amount of distortion is a non-linear f of distance from center of image

From ideal (u, v) to distorted pixel coordinates (u_d, v_d):

- Simple quadratic distortion model:

$$\begin{bmatrix} u_d \\ v_d \end{bmatrix} = (1 + k_1 r^2) \begin{bmatrix} u - u_0 \\ v - v_0 \end{bmatrix} + \begin{bmatrix} u_0 \\ v_0 \end{bmatrix}$$

where

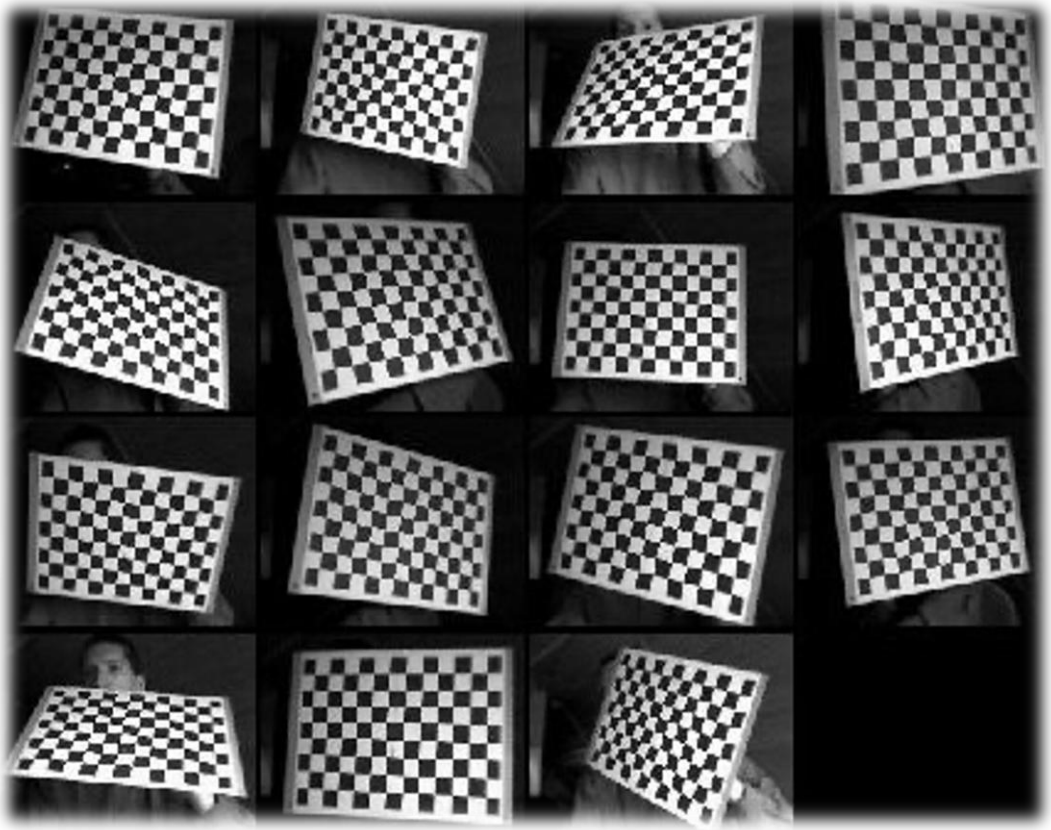
Radial Distortion parameter

$$r^2 = (u - u_0)^2 + (v - v_0)^2 .$$

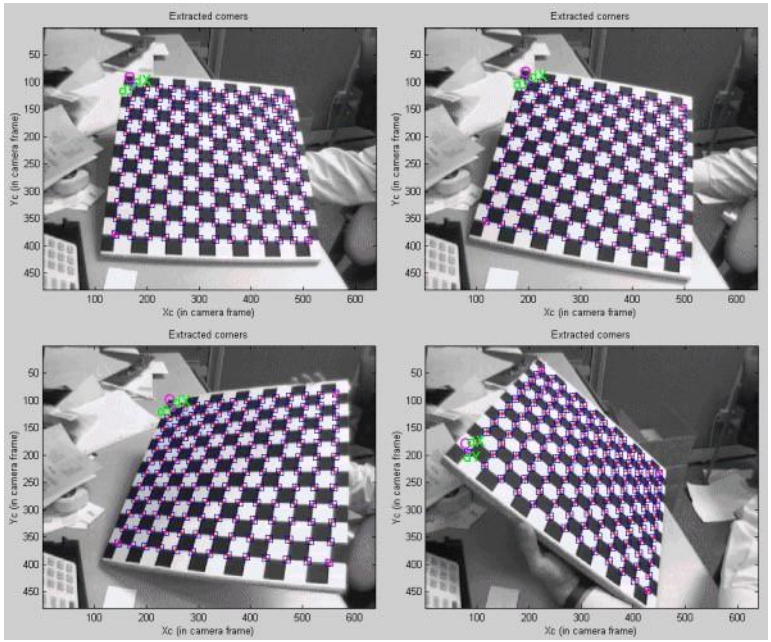
- Works well for most lenses

Camera Calibration

- Finding the *intrinsic parameters* of camera



Camera Calibration



- Use camera model to interpret the projection from world to image plane
- Using known correspondences of $\boldsymbol{p} \Rightarrow \boldsymbol{P}$, we can compute the unknown parameters K, R, $\boldsymbol{t}$ by applying the perspective projection equation
- so associate known, physical distances in the world to pixel-distances in image

$$\lambda \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = K[R \mid \boldsymbol{t}] \begin{bmatrix} X_W \\ Y_W \\ Z_W \\ 1 \end{bmatrix}$$

Projection matrix

How can we find out the values of the matrices *K*, *R*, ***t*** ?

Direct linear transform

- We know that : $\lambda \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = K[R \mid \boldsymbol{t}] \begin{bmatrix} X_W \\ Y_W \\ Z_W \\ 1 \end{bmatrix}$

- So there are 11 values to estimate: (the overall scale doesn't matter, so e.g. m_{34} could be set to 1)

$$\lambda \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \begin{bmatrix} m_{11} & m_{12} & m_{13} & m_{14} \\ m_{21} & m_{22} & m_{23} & m_{24} \\ m_{31} & m_{32} & m_{33} & m_{34} \\ m_{41} & m_{42} & m_{43} & m_{44} \end{bmatrix} \begin{bmatrix} X_W \\ Y_W \\ Z_W \\ 1 \end{bmatrix}$$

- Each observed point gives us a pair of equations

$$u_i = \frac{\lambda u_i}{\lambda} = \frac{m_{11}X_i + m_{12}Y_i + m_{13}Z_i + m_{14}}{m_{31} + m_{32} + m_{33} + m_{34}}$$

$$v_i = \frac{\lambda v_i}{\lambda} = \frac{m_{21}X_i + m_{22}Y_i + m_{23}Z_i + m_{24}}{m_{31} + m_{32} + m_{33} + m_{34}}$$

- To estimate 11 unknowns, we need **at least 6** points to calibrate the camera $\Rightarrow$ solved using linear least squares

Direct linear transform

$$\lambda \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \begin{bmatrix} m_{11} & m_{12} & m_{13} & m_{14} \\ m_{21} & m_{22} & m_{23} & m_{24} \\ m_{31} & m_{32} & m_{33} & m_{34} \\ m_{41} & m_{42} & m_{43} & m_{44} \end{bmatrix} \begin{bmatrix} X_W \\ Y_W \\ Z_W \\ 1 \end{bmatrix} = K[R \mid \boldsymbol{t}] \begin{bmatrix} X_W \\ Y_W \\ Z_W \\ 1 \end{bmatrix}$$

- **What we obtained**: the 3 x 4 projection matrix,
  what we need: its decomposition into the camera calibration matrix K, and the rotation R and position $\boldsymbol{t}$ of the camera.

- Use **QR factorization** to decompose the 3x3 submatrix (m_{11}:m_{33}) into the product of an upper triangular matrix K and a rotation matrix R (orthogonal matrix)

- The translation $\boldsymbol{t}$ can subsequently be obtained by:

$$\boldsymbol{t} = K^{-1} \begin{bmatrix} m_{14} \\ m_{24} \\ m_{34} \end{bmatrix}$$

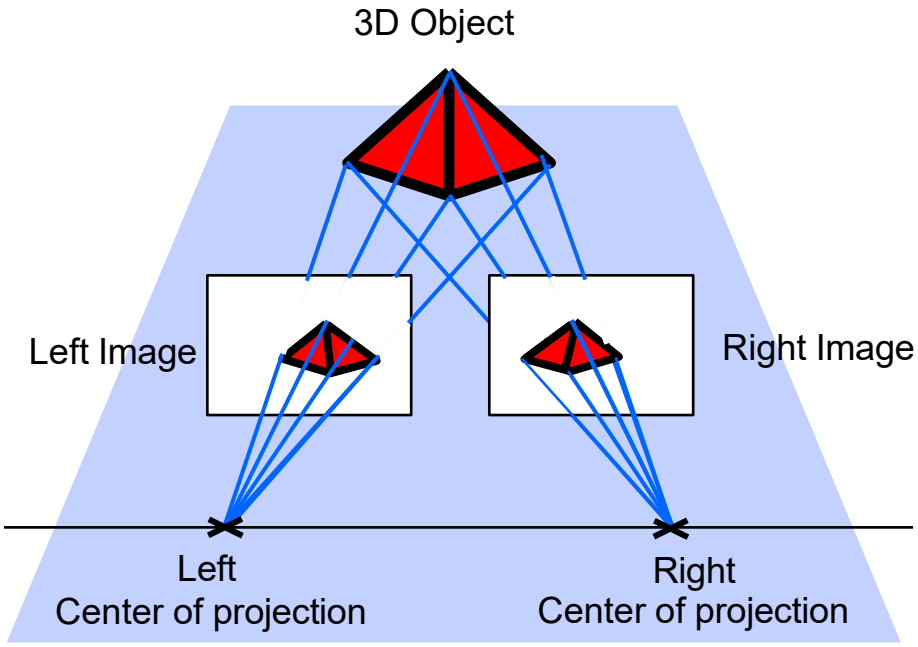
Next big question: How do we measure distances with cameras?





Camera distance estimation

- From a single image: we can only deduct the ray along which each image-point lies



- Observe the scene from 2 different viewpoints $\Rightarrow$ solve for the intersection of the rays and recover the 3D structure

Camera distance estimation methods

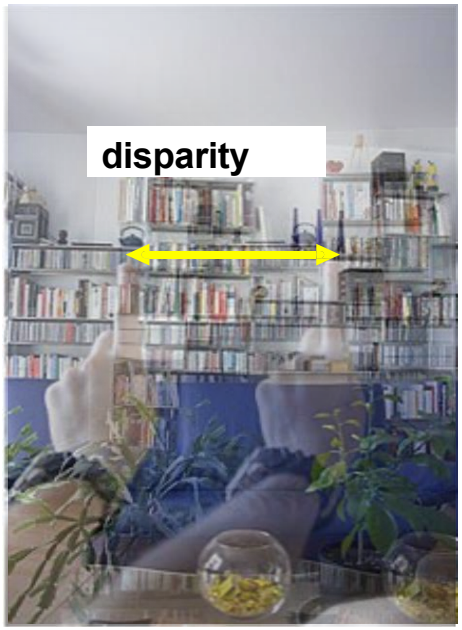
- **Stereo vision**: using 2 cameras with known relative position *T* and orientation *R*, recover the 3D scene information

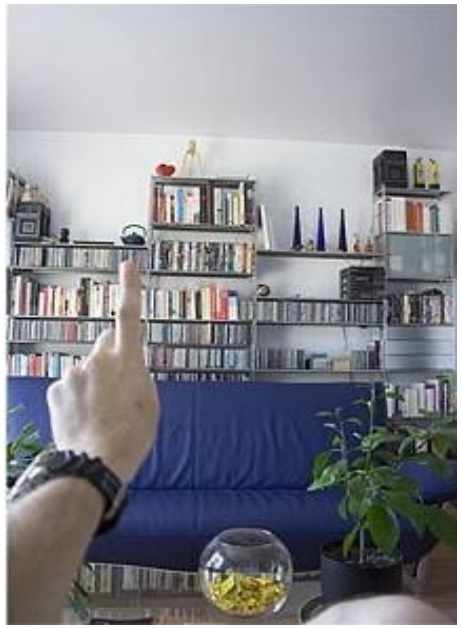


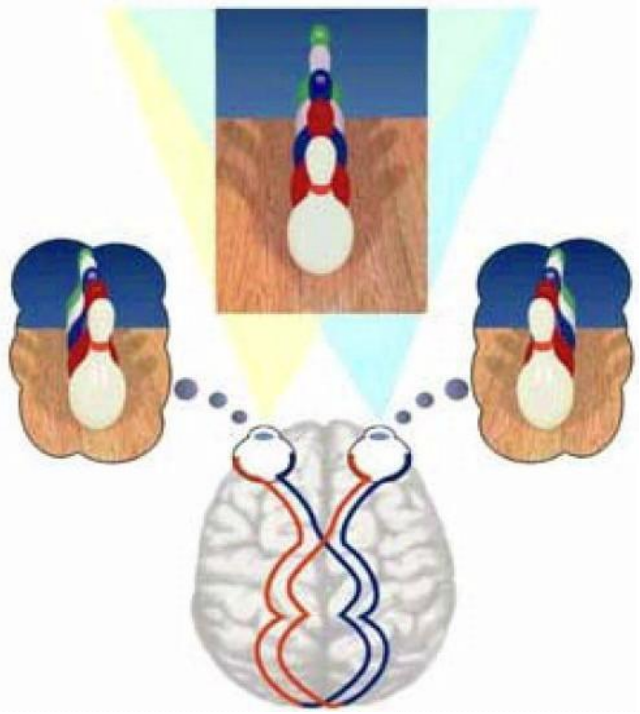
- **Structure from Motion**: recover the 3D scene structure and the camera poses (up to scale) from multiple images, from potentially unknown cameras

Disparity in the human retina

- **Stereopsys**: the brain allows us to see the left and right retinal images as a single 3D image
- Observe image disparity







Stereo Vision | basic case

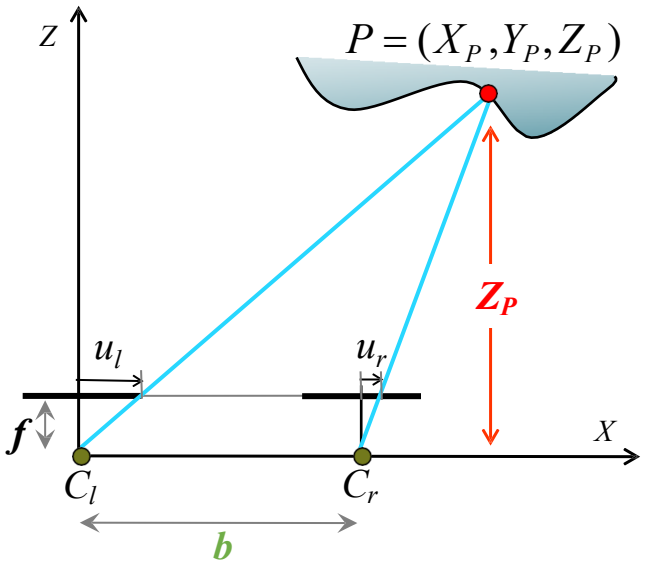
- An ideal, simplified case: assume both cameras are identical and are aligned with the x-axis

From Similar Triangles:

$$\frac{f}{Z_P} = \frac{u_l}{X_P}$$

$$\frac{f}{Z_P} = \frac{u_r}{X_P - b}$$

$$Z_P = \frac{bf}{u_l - u_r}$$



Disparity
difference in image location of the projection of a 3D point in two image planes

Baseline
distance between the optical centers of the two cameras

Disparity map

- **Disparity**: the difference in image location of corresponding 3D points as projected to the left and right images
- Disparity map holds the disparity value **at every pixel**
- Identify correspondent points of all image pixels in the original images
- Compute the disparity for each pair of correspondences
- Nearby objects experience bigger disparity:

$$Z_P = \frac{bf}{u_l - u_r} \Rightarrow u_l - u_r = \frac{bf}{Z_P}$$



Left image

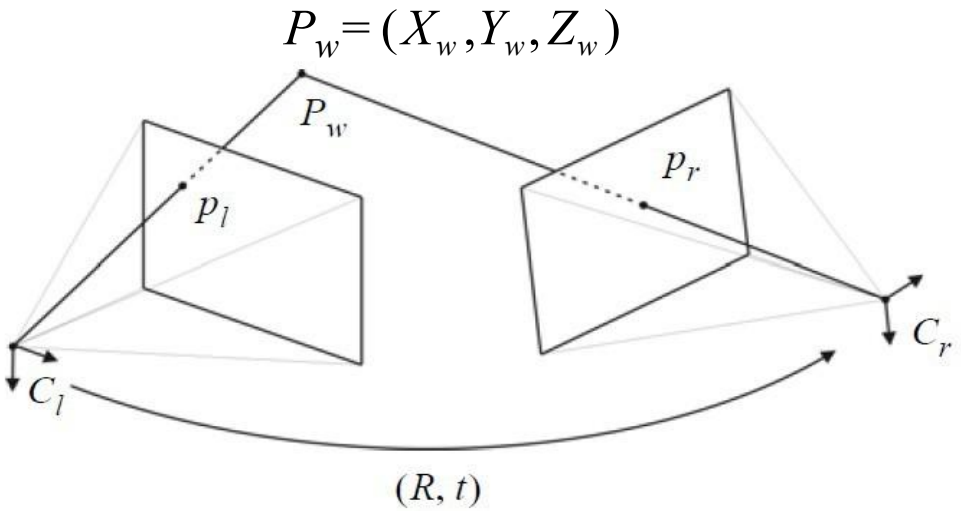


Right image



Stereo Vision: general case

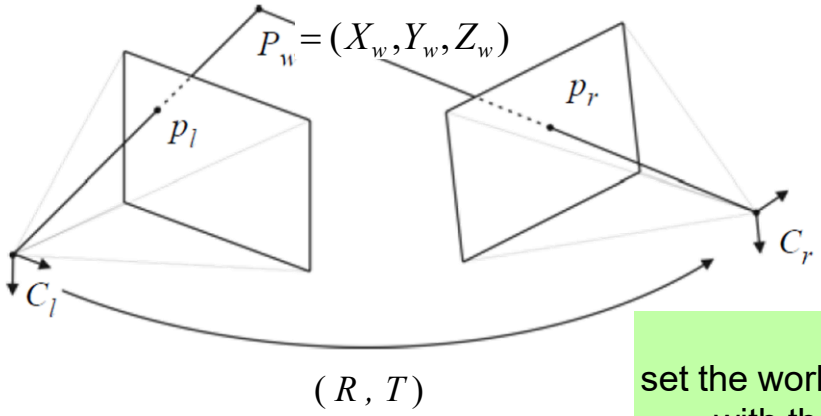
- Two identical cameras do not exist in nature!
- Aligning both cameras on a horizontal axis is very difficult
- In order to use a stereo camera, we need:
  - **Extrinsic parameters**: The relative pose between the cameras (rotation R, translation T)
  - **Intrinsic parameters**: The focal length f, optical center C, radial distortion of each camera





Stereo Vision: general case

- To estimate the 3D position of P_w we construct the system of equations of the left and right camera



Left camera:
set the world frame to coincide with the left camera frame

$$\tilde{p}_l = \lambda_l \begin{bmatrix} u_l \\ v_l \\ 1 \end{bmatrix} = K_l \begin{bmatrix} I & 0 \end{bmatrix} \begin{bmatrix} X_w \\ Y_w \\ Z_w \\ 1 \end{bmatrix}$$

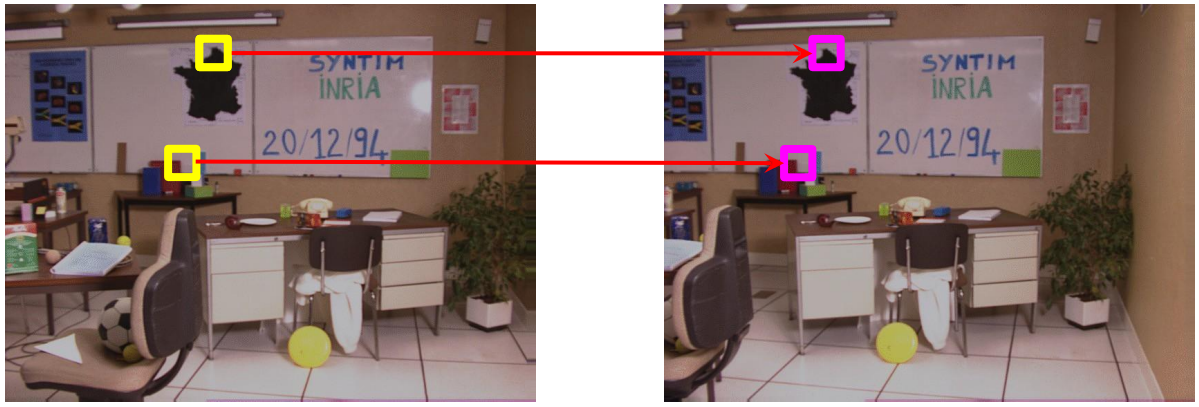
Right camera:
Rotation *R* and translation *T* from left camera

$$\tilde{p}_r = \lambda_r \begin{bmatrix} u_r \\ v_r \\ 1 \end{bmatrix} = K_r \begin{bmatrix} R & T \end{bmatrix} \begin{bmatrix} X_w \\ Y_w \\ Z_w \\ 1 \end{bmatrix}$$

- **Triangulation**: the problem of determining the 3D position of a point, given a set of corresponding image locations and known camera poses.

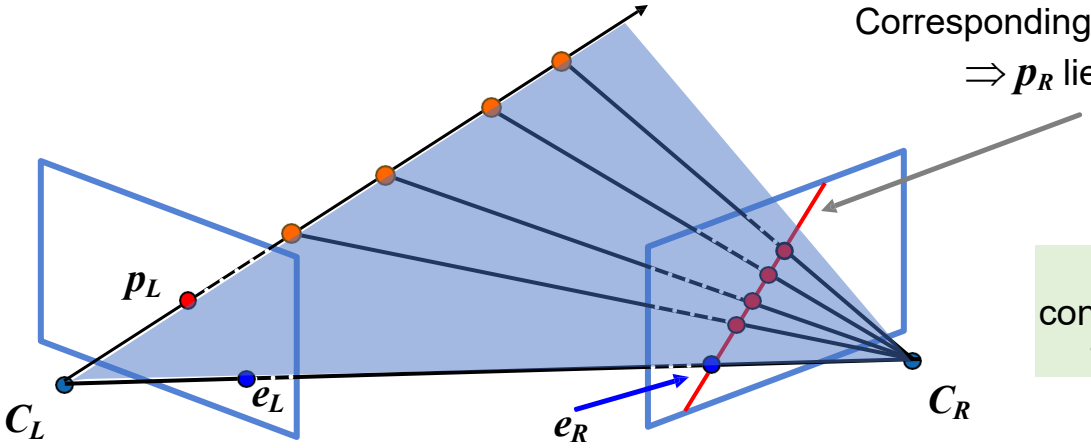
Correspondence search

- **Goal**: identify image regions / patches in the left & right images, corresponding to the same scene structure
  - **Typical similarity measures**: Normalized Cross-Correlation (NCC) , Sum of Squared Differences (SSD), Sum of Absolute Differences (SAD), …
  - Exhaustive image search can be computationally very expensive! Can we search for correspondences more efficiently?



The epipolar constraint

- Potential matches for p_L have to lie on the corresponding epipolar line
- **Epipolar line**: the projection of the infinite ray that connects C_L and p_L, onto the image plane of C_R (and vice versa)
- **Epipole**: the projection of the optical center C_L onto the image plane of C_R (and vice versa),i.e., e_L and e_R
- **Epipolar plane** is defined by p_L, C_L, C_R



Impose the epipolar constraint to aid the search for correspondences

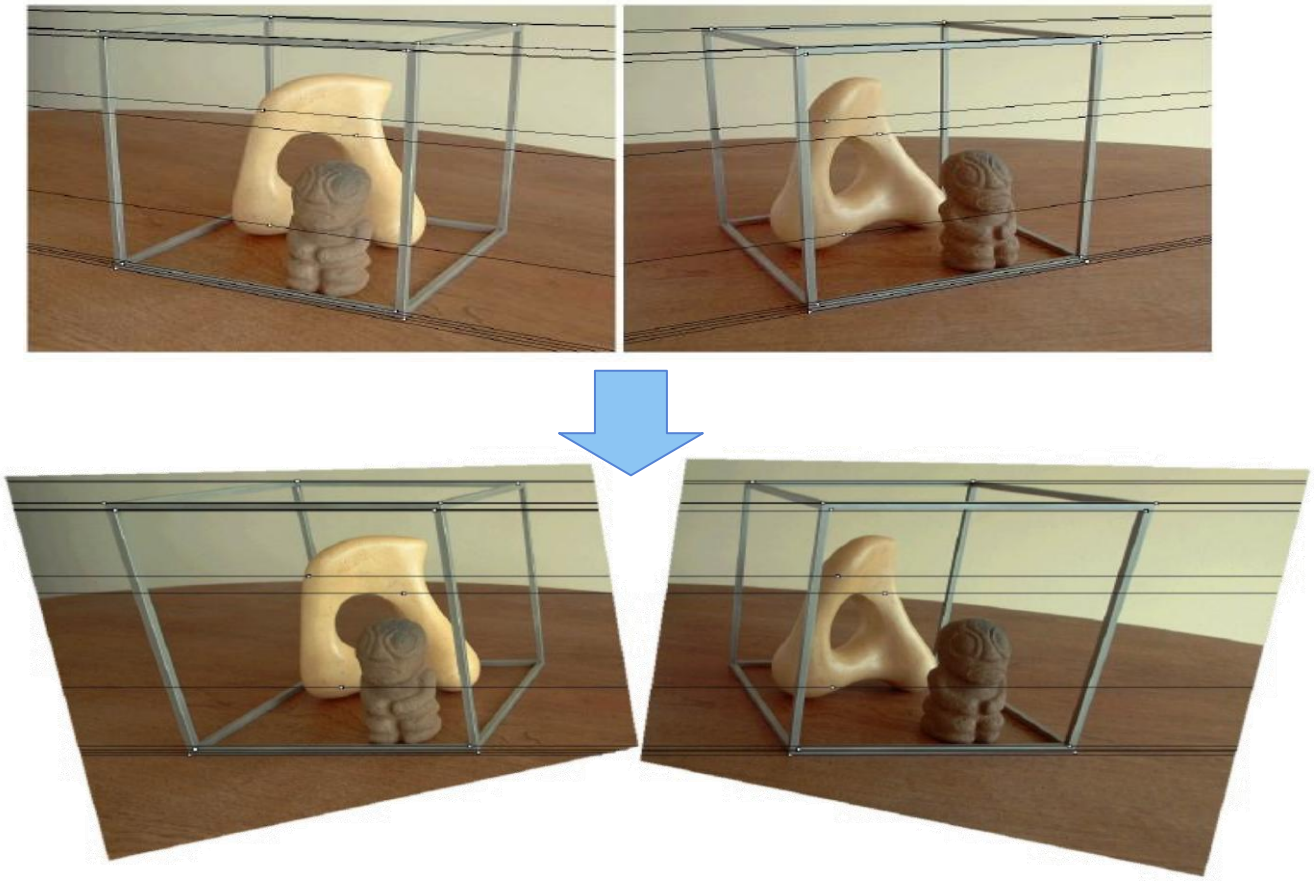
The epipolar constraint

- Thanks to the epipolar constraint, corresponding points can be searched for, **along epipolar lines**
  ⇒ computational cost reduced to 1 dimension!



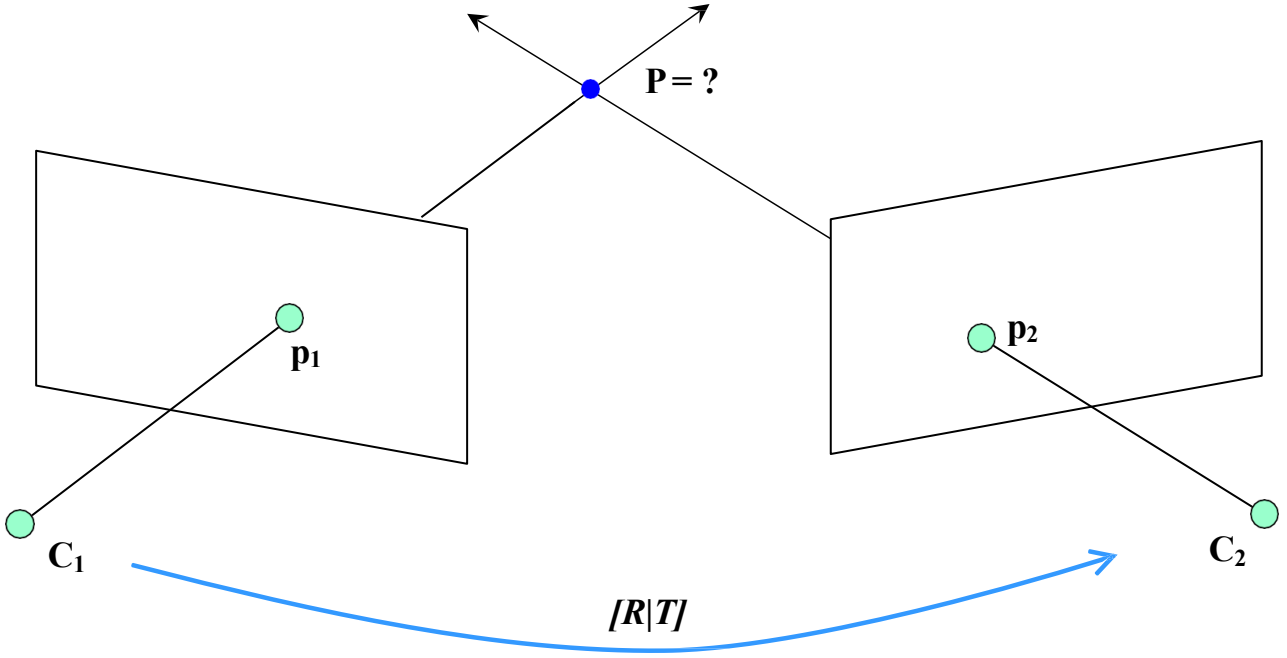
Epipolar Rectification: an example

- **Epipolar rectification**: Given a pair of stereo images, epipolar rectification is a transformation of each image plane such that all corresponding epipolar lines become collinear and parallel to one of the image axes, for convenience usually the **horizontal axis**.



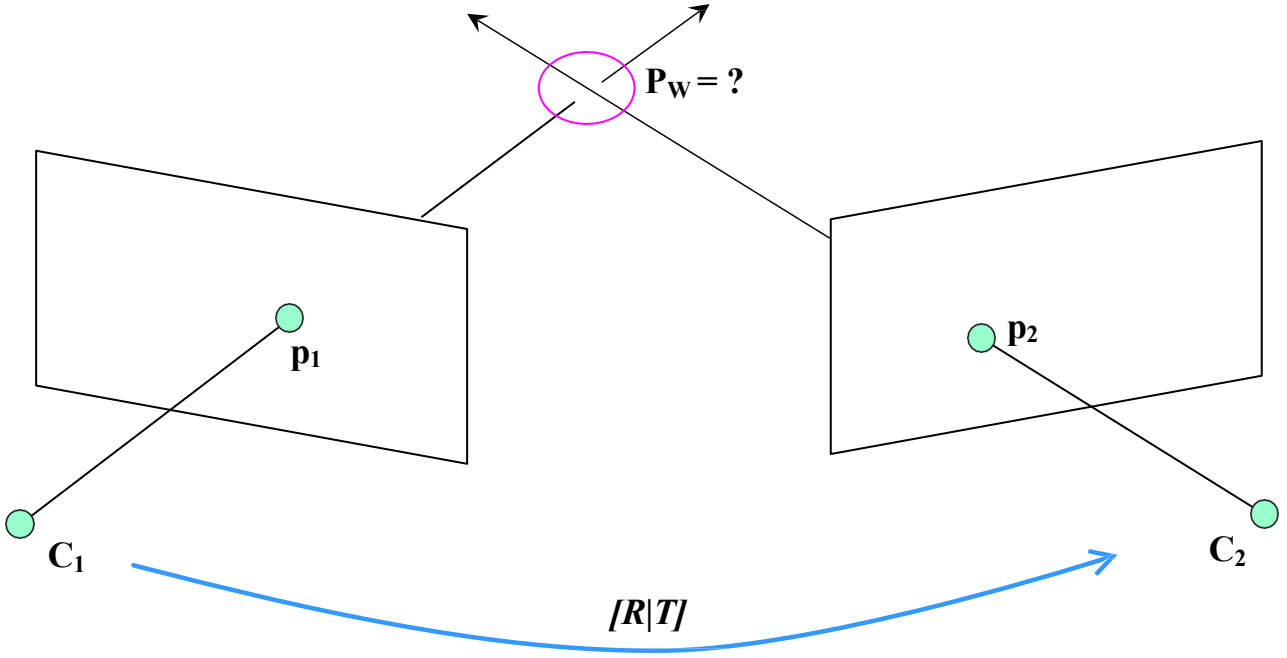
Triangulation

- **Triangulation**: Given the projections $\mathbf{p_1}$ and $\mathbf{p_2}$ of a 3D point $\mathbf{P}$ in two or more images (with known camera matrices $\boldsymbol{R}$ and $\boldsymbol{T}$), find the coordinates of the 3D point $\mathbf{P}$



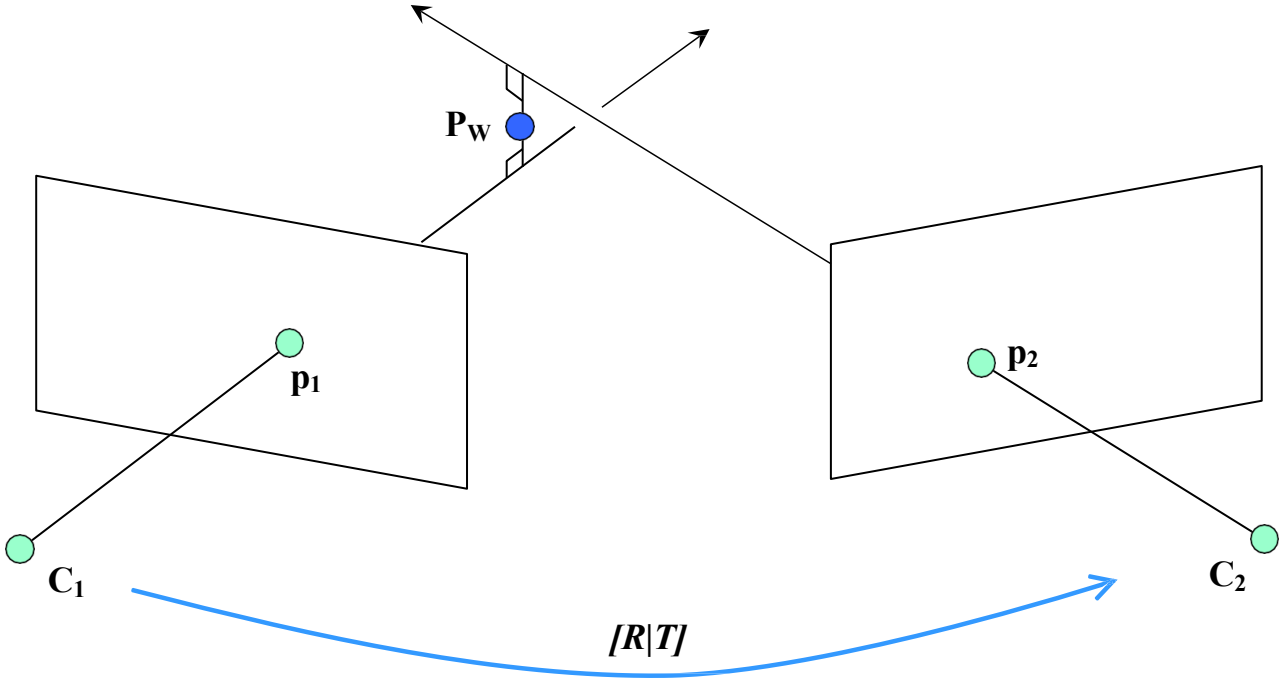
Triangulation

- We want to intersect the two visual rays corresponding to $\mathbf{p_1}$ and $\mathbf{p_2}$, but because of noise and numerical errors, they don't meet exactly



Triangulation: geometric approach

- Find shortest segment connecting the two viewing rays and let $\mathbf{P}_W$ be the midpoint of that segment



Triangulation: linear approach

- Left camera:

$$\tilde{P}_1 = M_1\tilde{P}_W \Rightarrow M_1\tilde{P}_W - \tilde{P}_1 = 0 \Rightarrow \tilde{P}_1 \times M_1\tilde{P}_W - \tilde{P}_1 \times \tilde{P}_1 = 0$$
$$\Rightarrow [\tilde{P}_{1\times}]M_1\tilde{P} = 0 \qquad \tilde{P}_1 \times \tilde{P}_1 = \mathbf{0}$$

- Right camera:

$$\tilde{P}_2 = M_2\tilde{P}_W \Rightarrow M_2\tilde{P}_W - \tilde{P}_2 = 0 \Rightarrow \tilde{P}_2 \times M_2\tilde{P}_W - \tilde{P}_2 \times \tilde{P}_2 = 0$$
$$\Rightarrow [\tilde{P}_{2\times}]M_2\tilde{P} = 0 \qquad \tilde{P}_2 \times \tilde{P}_2 = \mathbf{0}$$

- Cross product as matrix multiplication:

$$\boldsymbol{a} \times \boldsymbol{b} = \begin{bmatrix} 0 & -a_z & a_y \\ a_z & 0 & -a_x \\ -a_y & a_x & 0 \end{bmatrix} \begin{bmatrix} b_x \\ b_y \\ b_z \end{bmatrix} = [\boldsymbol{a}_\times]\boldsymbol{b}$$

Triangulation: linear approach

- Left camera:

$$\tilde{P}_1 = M_1\tilde{P}_W \Rightarrow M_1\tilde{P}_W - \tilde{P}_1 = 0 \Rightarrow \tilde{P}_1 \times M_1\tilde{P}_W - \tilde{P}_1 \times \tilde{P}_1 = 0$$
$$\Rightarrow [\tilde{P}_{1\times}]M_1\tilde{P} = 0 \qquad \tilde{P}_1 \times \tilde{P}_1 = \mathbf{0}$$

- Right camera:

$$\tilde{P}_2 = M_2\tilde{P}_W \Rightarrow M_2\tilde{P}_W - \tilde{P}_2 = 0 \Rightarrow \tilde{P}_2 \times M_2\tilde{P}_W - \tilde{P}_2 \times \tilde{P}_2 = 0$$
$$\Rightarrow [\tilde{P}_{2\times}]M_2\tilde{P} = 0 \qquad \tilde{P}_2 \times \tilde{P}_2 = \mathbf{0}$$

- Solve linear equalities:

$$[\tilde{P}_{1\times}]M_1\tilde{P}_W = 0$$
$$[\tilde{P}_{2\times}]M_2\tilde{P}_W = 0$$

3 unknowns () : $P_W = [X_W\ Y_W\ Z_W]^T$

6 independent equations

⇒ Solve system: $A \cdot P = 0$ using

Singular value decomposition (SVD)

Example

Give one 3D point seen by a left and right pinhole camera.
The image points are:

$$\tilde{P}_1 = \begin{bmatrix} u_1 \\ v_1 \\ 1 \end{bmatrix} = \begin{bmatrix} 0.25 \\ 0.1 \\ 1 \end{bmatrix}, \tilde{P}_2 = \begin{bmatrix} u_2 \\ v_2 \\ 1 \end{bmatrix} = \begin{bmatrix} 0.15 \\ 0.1 \\ 1 \end{bmatrix}$$

Use the camera matrices

$$M_1 = [I \quad 0] = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}, M_2 = [I \quad t] = \begin{bmatrix} 1 & 0 & 0 & 0.2 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

Try to obtain the 3D point $P_W = [X_W \; Y_W \; Z_W]^T$

Example

Solution: Since $\tilde{P}_1 \times M_1 \tilde{P}_W = 0$, one can obtain that $-X_W + 0.25Z_W = 0$ and $-Y_W + 0.1Z_W = 0$.

Similarly, since $\tilde{P}_2 \times M_2 \tilde{P}_W = 0$, one can obtain that $-X_W + 0.15Z_W + 0.2 = 0$ and $-Y_W + 0.1Z_W = 0$.

From the first and third equation, we have $0.25Z_W = 0.15Z_W + 0.2 = X_W$ Thus, $Z_W = 2$.

Based on the first and second equation, it yields $X_W = 0.25Z_W = 0.5$, and $Y_W = 0.1Z_W = 0.2$

$P_W = [X_W \; Y_W \; Z_W]^T = [0.5, 0.1, 0.2]^T$

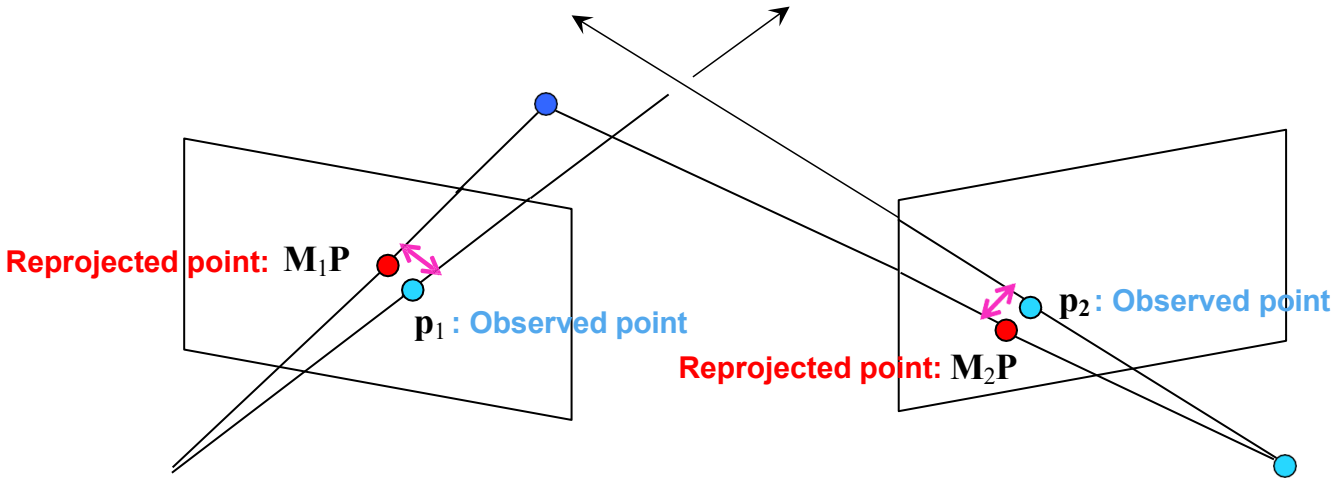
Triangulation: nonlinear approach

- Find $\boldsymbol{P}$ that minimizes the Sum of Squared **Reprojection Error**

$$SSRE = \|p_1 - M_1 P\|^2 + \|p_2 - M_2 P\|^2$$

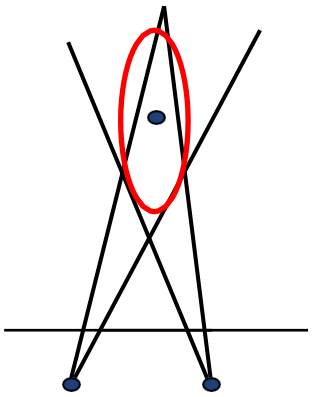
- In practice: initialize P using linear approach and then minimize SSRE using Gauss- Newton or Levenberg-Marquardt

$$P = ?$$

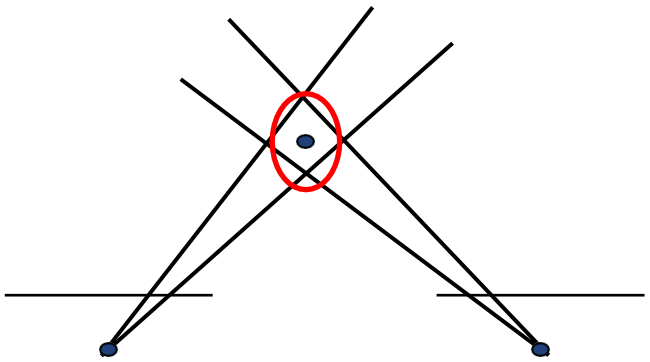


Sensitivity to baseline

- What is the optimal baseline **b**?
- Too small:
  - Large depth error
  - Can you quantify the error as a function of the disparity?
- Too large:
  - Object might be visible from one camera, but not the other
  - Difficult search problem for close objects



Small Baseline

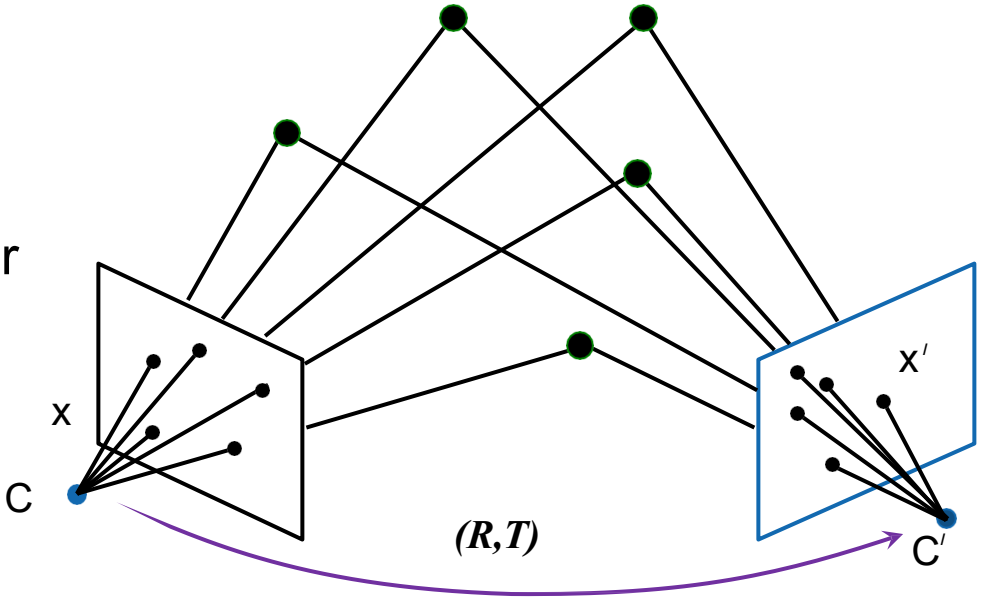


Large Baseline

SfM: Structure from Motion

- Given image point correspondences, $x_i, \ldots, x'$, determine both *R* and *T*.

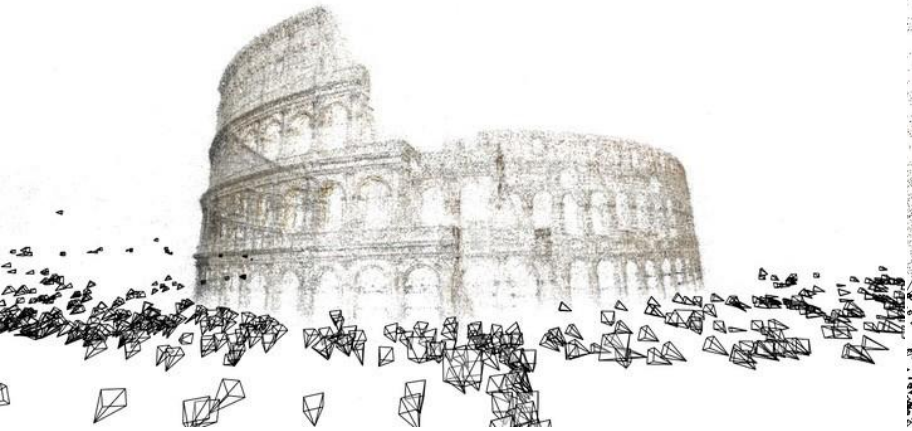
- Simultaneously estimate both 3D geometry (**structure**) and camera pose (**motion**): solve via non-linear minimization of the reprojection errors.



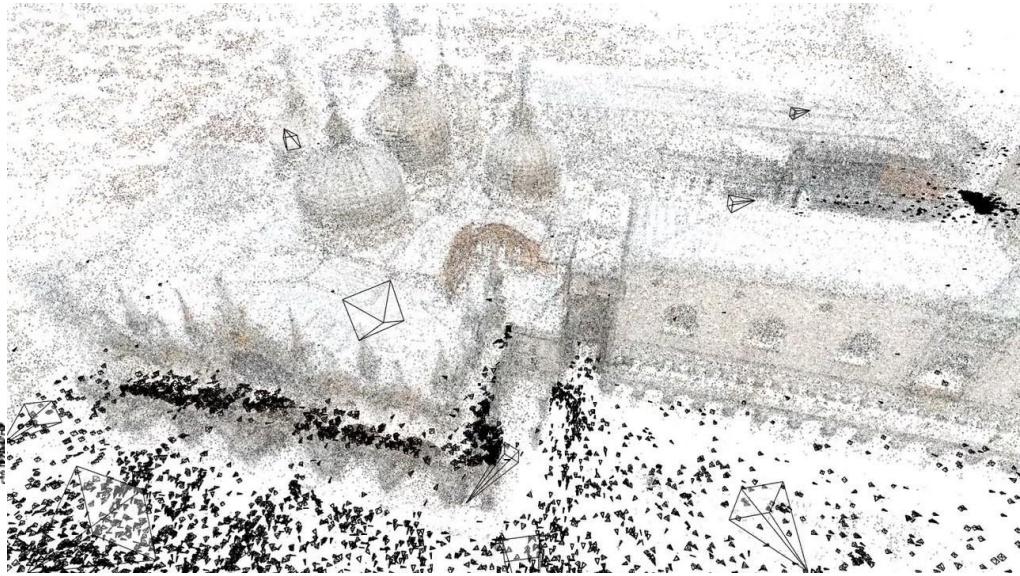


Multi-view SFM

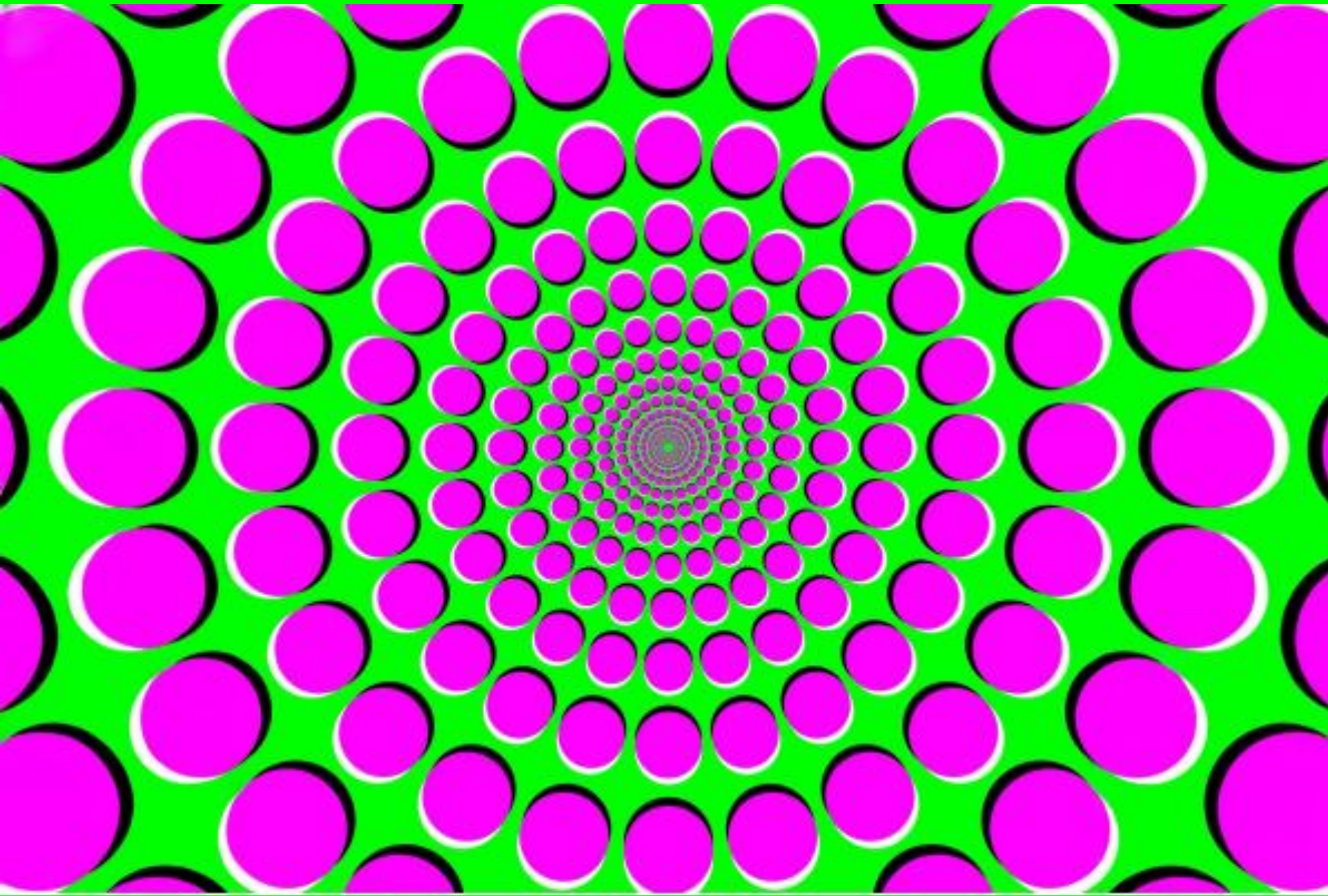
- Results of Structure from motion from user images from flickr.com
- Especially useful for mobile robots

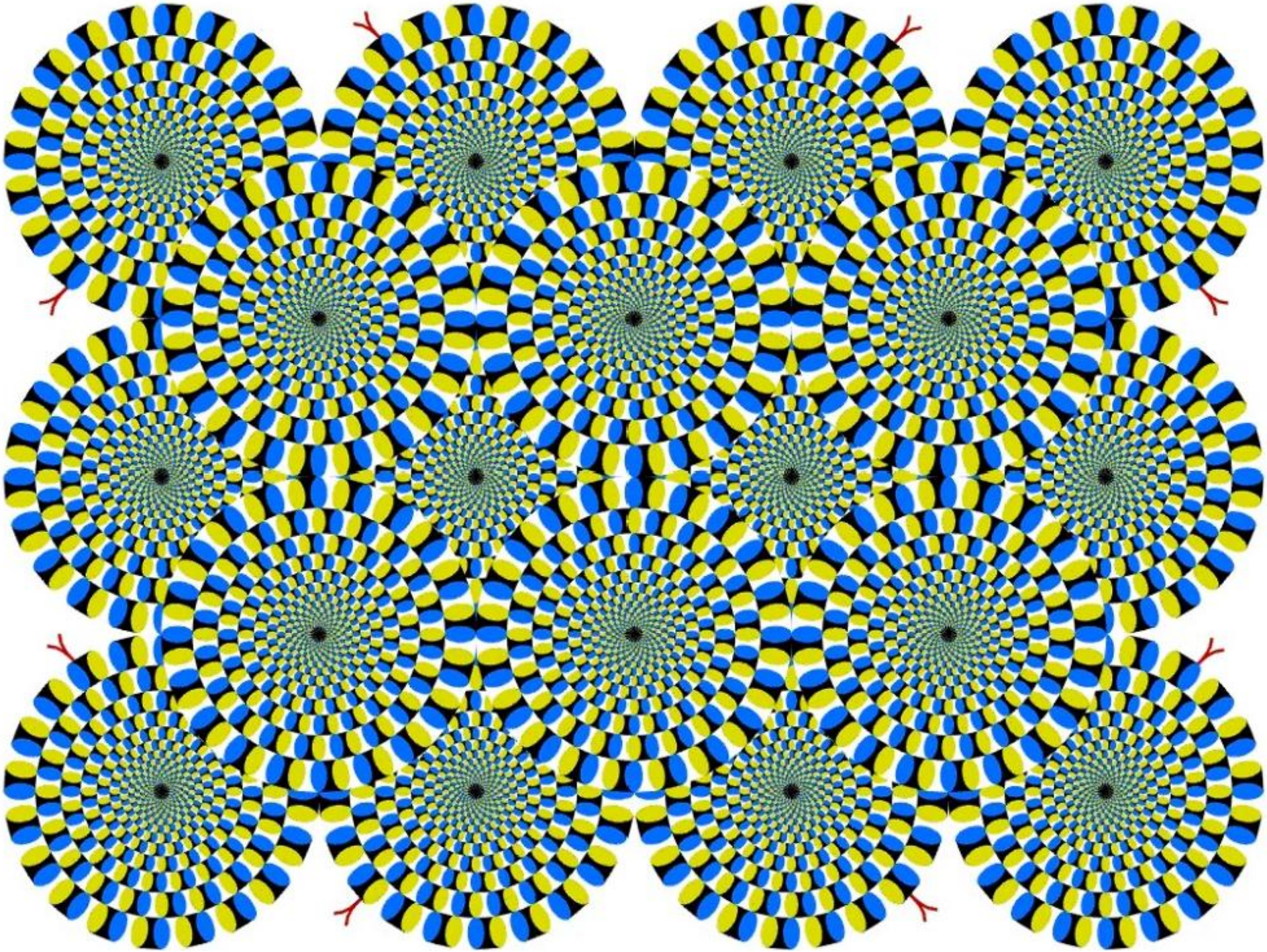


Colosseum, Rome
2,106 images, 819,242 points



San Marco square, Venice
14,079 images, 4,515,157 points





ROSE5730 Autonomous Mobile Robots

Lecture 4: Perception



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